

Chapter 63. Bronchodilator Therapy

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Bronchodilator Therapy: Introduction

Bronchodilators relax constricted airway smooth muscle in vitro. Because of this property, bronchodilators reverse airway obstruction, prevent bronchoconstriction and provide protection from constrictor stimuli.¹ In this chapter, bronchodilators employed in mechanically ventilated patients are discussed with a special emphasis on inhalation therapy.

Rationale

Ventilator-dependent patients with acute exacerbations of chronic obstructive pulmonary disease (COPD) and acute severe asthma routinely receive bronchodilators to relieve bronchoconstriction. By reducing airway resistance, bronchodilators reduce the pressure required to ventilate the lung. This reduction in pressure may protect the lung against injury and enhance patient comfort. A general population of ventilated patients in a medical intensive care unit (ICU)^{2,3} and patients with acute respiratory distress syndrome^{4,5} showed improvement in expiratory airflow and airway resistance after bronchodilators. Infants with bronchopulmonary dysplasia, and children with asthma, or bronchiolitis also receive bronchodilators on a routine basis.^{6–10} In ventilated patients with COPD, elevated airway resistance and intrinsic positive end-expiratory pressure are major causes for weaning failure.¹¹ In these patients, bronchodilators may facilitate weaning.¹² Therapy with bronchodilators is, therefore, routinely and commonly employed for many indications in ventilator-dependent patients.¹³

Pharmacologic Agents

β -adrenergic agonists, anticholinergic drugs, and methylxanthines are the three major classes of bronchodilators employed in the ICU. β -adrenergic agonists and anticholinergics are usually administered by the inhaled route, whereas methylxanthines can only be administered enterally or parenterally. Although they are not bronchodilators in the classic sense, corticosteroids, both inhaled and systemic, are commonly employed in acutely ill patients to reduce airway inflammation and increase airway caliber.^{14–16}

β -Agonists

The pharmacology of the β -agonists was extensively reviewed in the second edition of this textbook,¹⁷ and other excellent reviews are available.¹⁸

Route of Administration of β_2 -Adrenergic Agonists

β -agonists have been given by oral, subcutaneous, intravenous, and inhaled routes. Table 63-1 lists doses for individual drugs. Inhaled therapy is preferred because the drug is delivered directly to its site of action in the airways, a smaller quantity of drug produces an effect comparable to that observed with systemic administration, onset of effect is rapid, and systemic absorption of the drug is limited, thus minimizing side effects. The oral approach has been all but abandoned, and there appears to be no advantage to the intravenous route even in severe asthma with hypercapnia.¹⁹ A meta-analysis found no evidence of benefit for the intravenous use of β -agonists in patients who are refractory to inhaled β -agonists.²⁰

Table 63-1: Doses and Duration of Action of Commonly Used Bronchodilators in Ventilated Patients^a

Agents	Dose	Time Course (Onset, Peak, Duration)	Frequency of Dosing
Albuterol (Salbutamol)	SVN: 0.083% solution, 3 mL (2.5 mg), pMDI: 90 mcg/puff, 4 puffs	5 to 15 minutes, 30 to 60 minutes, 5 to 8 hours	4 to 6 times daily
Levalbuterol	SVN: 0.63 mg, or 1.25 mg	15 minutes, 30 to 60 minutes, 5 to 8 hours	3 to 4 times daily
Formoterol	SVN: 20 mcg/2 mL solution	1 to 3 minutes, 1 to 3 hours, 8 to 12 hours	2 times daily
Arformoterol	SVN: 15 mcg/2ml solution	1 to 3 minutes, 1 to 3 hours, 8 to 12 hours	2 times daily
Ipratropium	pMDI: 18 mcg/puff, 4 puffs SVN: 0.02% solution, 2.5 mL (0.5 mg)	15 minutes, 90 to 120 minutes, 6 to 8 hours	4 to 6 times daily
Albuterol + Ipratropium	pMDI: 90 mcg + 18 mcg/puff, 4 puffs SVN: 2.5 mg + 0.5 mg/dose	5 to 15 minutes, 30 to 60 minutes, 6 to 8 hours	4 to 6 times daily

Abbreviations: pMDI, pressurized metered-dose inhaler; SVN, small-volume nebulizer.

^aDry powder inhalers are not routinely employed in ventilated patients.

Individual Agents

Albuterol

Albuterol is today's standard short-acting bronchodilator. Its pharmacokinetics depend on the dose administered, the formulation of albuterol used (dry powder, pressurized metered-dose inhaler [pMDI], or nebulized) and the clinical situation (mechanically ventilated or ambulatory). Although increasing doses of albuterol produce greater bronchodilation, the optimum dose is difficult to predict. Peak bronchodilator response after 10 puffs of albuterol in ventilated patients was seen within 5 minutes and sustained for 60 minutes.¹² Duration of action of 6 puffs of albuterol delivered from a pMDI to ventilated patients ranged from less than 2 hours to more than 4 hours.²¹ Systemic effects are dose related, usually appearing within an hour of administration with a return to baseline within 4 hours.²²

Levalbuterol

Commonly available racemic albuterol is a 50-50 mixture of R-albuterol and S-albuterol. Levalbuterol, the R-enantiomer, was formulated to avoid possible adverse effects of the S-enantiomer. For doses that produce similar increases in forced expiratory volume in 1 second (FEV₁), increases in heart rate are less with levalbuterol (2 to 4 beats/min) than with albuterol,²³ and the bronchodilator effect is greater in patients with acute asthma.^{24,25} In patients receiving long-term mechanical ventilation, levalbuterol produced an increase in secretion volume but the effect dissipated rapidly.²⁶

Salmeterol

Salmeterol is a long-acting β_2 -agonist. Its onset of action is slower than that of albuterol in vitro (mean: 6.4 vs. 1.9 minutes)²⁷ and in vivo (mean: 10 vs. 4 minutes).²⁸ With salmeterol, peak bronchodilator response was seen at 5 hours versus 1 hour with albuterol, and FEV₁ was higher than predose FEV₁ levels for 12 hours.²⁸ Systemic effects of salmeterol are dose related, manifest later, and last longer; maximum effects on heart rate occur within 75 to 135 minutes and are still evident after 4 hours.²² In ventilated patients with acute exacerbations of COPD, four doses of salmeterol by pMDI (100 mcg) produced a bronchodilator effect within 30 minutes that was maintained for approximately 8 hours, with significant variability in the magnitude and duration of the response among patients.²⁹

Formoterol

Formoterol is a long-acting agent like salmeterol but is distinguished by its relatively quicker onset of action.³⁰ Duration of effect is approximately 12 hours. Because of its rapid onset of effect, formoterol is effective during acute exacerbations.³¹ Stereoselective formoterol (arformoterol) is available as a nebulized solution.

Once-Daily Long-Acting β -Agonists

Indacaterol, vilanterol, and carmoterol have duration of action of 24 hours or more and are in various stages of clinical development.³²

Anticholinergics

Anticholinergic Action and Airway Smooth Muscle Relaxation

The principal mechanism of action of muscarinic antagonists is to block vagally mediated bronchoconstriction. In patients with COPD, a greater bronchodilator response occurs after [atropine](#) than after albuterol. The bronchodilator response to [atropine](#) in patients with COPD is greater than that in normal subjects. These data suggest enhanced parasympathetic activity in COPD.^{33,34} Patients with chronic asthma do not show as much benefit with anticholinergic agents, although patients with acute asthma exacerbations benefit from inhaled anticholinergic agents.

Other Effects of Anticholinergic Action in the Lungs

The quaternary ammonium-containing muscarinic antagonists in current use have no significant effects on mucus secretion or the rheologic properties of mucus,³⁵ although some reports do suggest a decrease in sputum volume.³⁶ Mucociliary clearance is unchanged with inhaled ipratropium.^{37,38}

Acetylcholine can stimulate alveolar macrophages to release chemotactic factors.³⁹ Long-acting anticholinergics may thus decrease airway inflammation by blocking the actions of acetylcholine.⁴⁰

Side Effects of Anticholinergics

Although muscarinic receptors are widespread in the body, the quaternary ammonium group in modern antimuscarinic agents ([ipratropium](#) and [tiotropium](#)) results in poor systemic absorption allowing higher doses to be inhaled with few systemic effects.

Dryness of the mouth is a common side effect of anticholinergics, whereas blurring of vision occurs infrequently. Unilateral mydriasis is believed to result from direct contact of the nebulized solution with the eyes.^{41,42} [Ipratropium](#) inhalation does not worsen arterial oxygenation,⁴³ and tolerance to the bronchodilator or bronchoprotective effects of anticholinergics does not occur. Systemic side effects (tachycardia, palpitations, urinary hesitancy, constipation, blurred vision, or glaucoma) are more common with tertiary ammonium compounds, such as [atropine](#), than with [ipratropium](#) or [tiotropium](#). In the Lung Health Study, Anthonisen et al found a higher incidence of supraventricular arrhythmias as well as significant increases in hospitalization and mortality rates in the [ipratropium](#) group as compared to the placebo group.⁴⁴ A slightly higher risk of developing cerebrovascular accidents was suggested for patients receiving tiotropium (<http://www.fda.gov.wrnmmc.idm.oclc.org/medwatch/safety/2008/safety08.htm#Tiotropium>), and a metaanalysis found a higher risk of cardiovascular events in patients with COPD receiving [ipratropium](#) or [tiotropium](#),⁴⁵ although these have not been observed in a recently concluded large randomized controlled trial.⁴⁶ Nevertheless, [ipratropium](#) and [tiotropium](#) should be used with caution in patients with renal failure, prostatic hypertrophy, urinary retention, or glaucoma.

Individual Agents

Ipratropium

The usual dose is 36 mcg (2 puffs from a pMDI) every 4 to 6 hours. There is a plateau in the achievable improvement in FEV₁ with increasing doses with no significant improvement beyond a dose of 72 mcg.⁴⁷ In patients with severe airway obstruction, however, the amount of drug delivered to the site of action may vary and higher doses may be needed. The usual nebulized dose is 500 mcg every 6 hours. The onset of action is within 30 minutes of an inhaled dose and the effects last for about 4 to 6 hours.

Tiotropium

Tiotropium is a long-acting bronchodilator that has equal binding affinity for M₁, M₂, and M₃ receptor subtypes. It dissociates much more slowly from M₃ receptors than from M₂ receptors.⁴⁸ In view of the role of M₂ receptors in inhibiting acetylcholine release, this may represent a beneficial aspect of its action. Tiotropium is inhaled once daily. Onset of action is within 30 minutes. An increase in FEV₁ from baseline is present at 24 hours.⁴⁹

Corticosteroids

Corticosteroids have no direct action on contraction of airway smooth muscle; hence, they are not bronchodilators in the classic sense. Corticosteroids, however, offer bronchoprotection, and they improve airway obstruction by reducing airway inflammation and via vasoconstrictor effects.^{16,50}

Corticosteroids reduce airway inflammation by several mechanisms,^{51–54} decrease airway hyperresponsiveness and reduce the predilection to acute episodes of airflow obstruction. Corticosteroids also upregulate β -receptor expression on cell membranes, increase the proportion of β receptors in a high-affinity binding state, and inhibit the release of inflammatory mediators such as phospholipase A₂, which could destabilize membrane support of the β receptor.⁵⁵

Several corticosteroids, such as hydrocortisone, cortisone, prednisone, prednisolone, and methylprednisolone are given parenterally or orally for treatment of inflammation in asthma and COPD. Agents with high topical activity are employed as inhaled corticosteroids (ICSs), such as beclomethasone, triamcinolone, flunisolide, fluticasone, budesonide, mometasone, and ciclesonide, because they produce direct antiinflammatory effects while minimizing unwanted systemic side effects.

Systemic administration of corticosteroids is associated with several adverse side effects.⁵⁶ The use of ICSs is attractive mainly to reduce the side effects observed with systemic therapy.^{57,58} Side effects of therapy with ICSs include oropharyngeal fungal infections, hoarseness, hypothalamic–pituitary–adrenal axis suppression, reduction in bone mass, growth restriction in children, skin bruising, and increased propensity to develop pneumonia.^{57–60}

Clinical Use of Bronchodilator Drugs

Patients presenting with acute severe asthma and acute exacerbations of COPD are in urgent need of bronchodilation. Inhalation is the preferred route of administration of bronchodilator agents because it produces a rapid onset of action, requires smaller doses than those needed orally or parenterally, and minimizes systemic side effects.⁶¹ For administration of short-acting β -agonists the use of pMDI and holding chamber (a chamber spacer that incorporates a valve, so that aerosol is retained within the chamber for a finite time after pMDI actuation) is a convenient and cost-effective option and provides efficacy similar to that achieved with a jet nebulizer.⁶²

β -Agonists in Asthma and Chronic Obstructive Pulmonary Disease

In general, repeated administration of short-acting β -agonists at frequent intervals is recommended for relief of symptoms in patients with acute asthma.^{50,63} There is a dose–response relationship with increasing doses of β -agonists resulting in increasing bronchodilation.^{64,65} Rodrigo and Rodrigo, however, noted that 2.4 mg of albuterol every hour (400 mcg every 10 minutes) delivered via a pMDI/holding chamber produced clinically significant bronchodilation, and higher doses increased the incidence of adverse effects without enhancing bronchodilation.^{66,67} Other investigators have concluded that the cumulative dose, and not the dosage regimen, influences the bronchodilator response (i.e., the same response was obtained when 5 mg was given as a single dose as when it was given as two doses of 2.5 mg).^{68,69} The usual dose necessary to produce bronchodilation in severe asthma is between 5 and 10 mg of albuterol.⁷⁰ Use of higher doses during the initial phase of an acute exacerbation of asthma may have clinical utility. A faster response may be seen, and, because most patients respond to 5 to 10 mg of albuterol, nonresponders could be identified. Nonresponders, defined as those requiring admission, not discharge, to an emergency department, were found to have a flatter dose–response curve to albuterol, with peak expiratory flow remaining below 45% despite administration of high doses of albuterol.^{68,71,72}

The logistical problem in repeatedly administering doses of nebulized albuterol (intermittent therapy) stimulated interest in continuous nebulization. Various investigators have reported conflicting findings with continuous nebulization.^{73,74} Two meta-analysis of published trials also reached different conclusions.^{75,76} Overall, it appears that patients with more severe airway obstruction could benefit from continuous nebulization,^{4,77} especially if they do not respond to therapy within the first hour in the emergency department.

Both formoterol and arformoterol are effective for treatment of patients with COPD.^{78–81} There is not much published experience with their use during acute exacerbations of asthma or COPD.

In summary, the severity of an individual episode is better measured as response to bronchodilator therapy rather than by baseline pulmonary function. Continuous nebulization may be employed in patients with acute severe asthma in whom pulmonary function does not improve by the end of the first hour of intensive bronchodilator therapy.

Choice of Delivery System

In clinical studies evaluating bronchodilator administration through pMDI/holding chamber versus nebulizers, equivalent changes in FEV₁ were produced by 6 mg/hour of albuterol given by the nebulized route and 2.4 mg/hour of albuterol given by pMDI/holding chamber, an equivalent dose ratio

of 2.5:1. A higher incidence of tremor and anxiety and higher serum albuterol was seen in the nebulizer group.⁸² Improvement in pulmonary function is similar^{70,83,84} or slightly better^{85,86} with pMDI/holding chamber than with nebulizers, while increases in heart rate are more frequent with nebulizers.⁷⁰

Anticholinergics in Asthma and Chronic Obstructive Pulmonary Disease

The ideal bronchodilator and optimal regimen for acute exacerbations of COPD have not been established. A pooled analysis comparing the effects of β -agonists (fenoterol and metaproterenol) and anticholinergics (*ipratropium*) found no significant differences in improvement in FEV₁.⁸⁷ If response to β -agonists is suboptimal or the exacerbation is severe, *ipratropium* can be added.⁸⁸

Corticosteroids in Asthma and Chronic Obstructive Pulmonary Disease

A short course of systemic corticosteroids may be necessary to gain control of asthma symptoms. ICS may be started at the same time and the systemic steroid tapered slowly. Subsequent adjustments in ICS dose are based on assessments of symptoms, rescue use of short-acting β -agonists, and measurement of peak expiratory flow rates.

High-dose ICSs administered by pMDI to patients with acute asthma exacerbations achieve similar improvements in symptoms and pulmonary function as systemically administered corticosteroids.^{89–91}

Corticosteroids for Chronic Obstructive Pulmonary Disease

In patients with COPD, regular use of ICSs produces a small increase in pulmonary function and improves health status.^{92–94} ICSs, however, do not alter the rate of decline in FEV₁ and they do not reduce mortality.⁹⁵ In acute exacerbations, oral or parenteral corticosteroids are often employed with some benefit for patients who are admitted to the hospital.^{96,97} Current evidence favors moderate over high doses, and short courses of oral corticosteroids over intravenous therapy for treating acute exacerbations.^{98–100}

Risk-to-benefit considerations have prompted investigation of ICS use for treatment of acute exacerbations of COPD. Several investigators have shown that the efficacy of nebulized *budesonide* is comparable to that observed with systemic corticosteroids, except in the most severely ill patients.^{101–104}

In patients receiving long-term ventilation for severe COPD, Nava et al observed a small, but statistically significant, reduction in airway resistance with fluticasone.¹⁰⁵ The optimal methods of administering ICS and the appropriate doses in ventilated patients have not been determined, but are likely to be higher than those used for maintenance therapy in COPD.^{16,101–104} Moreover, many ventilated patients receive systemic corticosteroids for acute exacerbations of asthma and COPD, and it is unclear if patients derive additional benefits from ICSs in the presence of high-dose therapy with oral or parenteral corticosteroids. Finally, regular use of ICSs in ambulatory patients with COPD is associated with a higher risk of pneumonia.⁶⁰ Because ventilated patients are already vulnerable to developing pneumonia for a variety of reasons, the potential to add another risk factor by administering ICSs requires serious consideration. In summary, ICSs may have a limited role in ventilated patients, but further investigations are needed to determine the appropriate dosing regimen and risk-to-benefit ratio of using ICSs in this group of patients.

Combination Therapy

β -Agonists and Anticholinergics

In acute asthma, some investigators found additional benefits with combined anticholinergic and β -agonist therapy over β -agonists alone,^{106–108} whereas others^{109,110} found no difference. A meta-analysis that compared albuterol alone to albuterol with *ipratropium* found an overall improvement of 7.3% in FEV₁ and 22.1% in peak expiratory flow rates, and reduction in hospitalization rates for studies in which data were available.¹¹¹ A more recent meta-analysis by Rodrigo and Rodrigo¹¹² found benefit for a combination of anticholinergics with albuterol in severe asthma (FEV₁ <50%).

In patients with acute exacerbations of COPD, adding an anticholinergic to a β -agonist may¹¹³ or may not^{114–116} provide greater bronchodilation. Fernandez et al¹¹⁷ found an improvement in airway pressures in ventilated patients with the combination of *ipratropium* and fenoterol in comparison to *ipratropium*; *ipratropium* alone achieved no benefit.

Bronchodilators and Inhaled Corticosteroids

There is little doubt that combinations of ICSs plus long-acting β -agonists (Advair Diskus; Advair pMDI; Symbicort pMDI) are the most effective and widely used treatments currently available for treatment of asthma. A combination of formoterol and [budesonide](#), used as a reliever as well as maintenance therapy, may be employed for treatment of mild to moderate acute asthma exacerbations that could be managed at home. This management strategy significantly reduces the frequency of more severe exacerbations.¹¹⁸ Currently, there is limited experience with bronchodilator and ICS combination therapy for treatment of acute exacerbations of COPD, especially in ventilated patients.

Methylxanthines

Theophylline is an inexpensive and commonly employed bronchodilator in many developing countries. The reader is referred to the second edition of this book¹⁷ for details on its pharmacology and mechanisms of action.

Theophylline administration requires empiric loading and maintenance dosing with frequent measurement of serum levels and dose adjustment.¹¹⁹ Plasma theophylline levels should be maintained between 5 and 15 mg/L because adverse effects frequently occur with higher levels.^{120–122} At concentrations less than 15 mg/L, theophylline is a relatively weak bronchodilator and its beneficial effects are more likely explained by its antiinflammatory action, particularly in synergy with corticosteroids.¹²³

In acute asthma, theophylline does not confer additional bronchodilation in patients receiving intensive therapy with inhaled β -agonists and intravenous corticosteroids.^{124–126} Likewise, routine use of theophylline in acute exacerbations of COPD is not supported by randomized controlled trials.¹¹⁴ Theophylline may have a role in reducing hospitalization among patients receiving emergency department treatment for asthma and COPD,¹²⁷ and in promoting corticosteroid activity in patients with acute exacerbations of COPD.¹²⁸ Few investigators have reported on the use of theophylline in ventilated patients.^{117,129,130} In the neonatal ICU, theophylline is employed to prevent apneas.¹³¹ In adults, loading doses of intravenous theophylline produced significant bronchodilation, comparable to that achieved with two inhalations of either albuterol or [ipratropium](#) bromide.^{117,129,130} The increased respiratory muscle strength with theophylline^{132–134} may help in weaning.

Frequent side effects with theophylline, especially nausea and vomiting, are a major drawback. Older patients and those with a low serum albumin are particularly susceptible to serious theophylline toxicity, such as cardiac arrhythmias and seizures.^{135–137} In summary, the high frequency of side effects, its relatively low efficacy, frequent drug interactions, complicated dosing regimens, and the need for repeatedly monitoring serum levels have significantly limited theophylline use in the ICU.

Selective Phosphodiesterase E₄ Inhibitors

Phosphodiesterase E₄ (PDE₄) is the predominant isozyme responsible for metabolizing cyclic adenosine monophosphate in airway smooth muscle, and in immune and inflammatory cells.^{138,139} Elevation of cyclic adenosine monophosphate by selective PDE₄ inhibition has a wide variety of pharmacologic effects. Orally administered selective PDE₄ inhibitors have bronchodilator and antiinflammatory effects in patients with asthma and COPD.^{140–145} Unlike theophylline, these agents do not exhibit significant drug interactions.^{146,147} Side effects, such as nausea and headache, are generally mild to moderate, but could limit the use of these agents in critically ill patients.

Leukotriene-Modifying Agents

Leukotrienes are formed by the action of 5-lipoxygenase on cell membrane-derived arachidonic acid. Zileuton blocks 5-lipoxygenase and inhibits leukotriene synthesis, whereas zafirlukast, montelukast, and pranlukast all block the final step in the action of leukotrienes on the leukotriene receptor. The leukotrienes are mainly employed in patients with chronic asthma as adjuncts to ICSs. In acute asthma, intravenous montelukast achieves modest early gains in lung function without significant improvement in rates of hospital admission.¹⁴⁸

Factors Influencing Aerosol Delivery during Mechanical Ventilation

Several factors influence aerosol delivery during mechanical ventilation, including variables related to the aerosol-generating device, the ventilator and ventilator circuit, the inhaled drug or agent, and the patient ([Fig. 63-1](#)). Aerosol delivery during mechanical ventilation depends to a great extent on the type of aerosol-generating device employed.

Figure 63-1

Ventilator-related

- Ventilation mode
- Tidal volume
- Respiratory rate
- Duty cycle
- Inspiratory waveform
- Breath-triggering mechanism



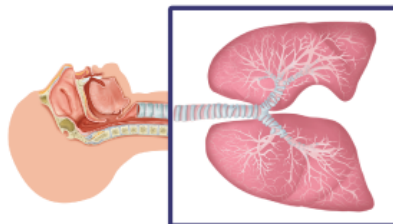
Device-related—MDI

- Type of spacer or adapter
- Position of spacer in circuit
- Timing of MDI actuation
- Type of MDI



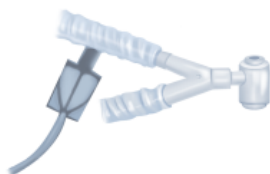
Drug-related

- Dose
- Formulation
- Aerosol particle size
- Targeted site for delivery
- Duration of action



Device-related—nebulizer

- Type of nebulizer
- Fill volume
- Gas flow
- Cycling: inspiration vs. continuous
- Duration of nebulization
- Position in the circuit



Circuit-related

- Endotracheal tube size
- Humidity of inhaled gas
- Density of inhaled gas

Patient-related

- Severity of airway obstruction
- Mechanism of airway obstruction
- Presence of dynamic hyperinflation
- Patient-ventilator synchrony

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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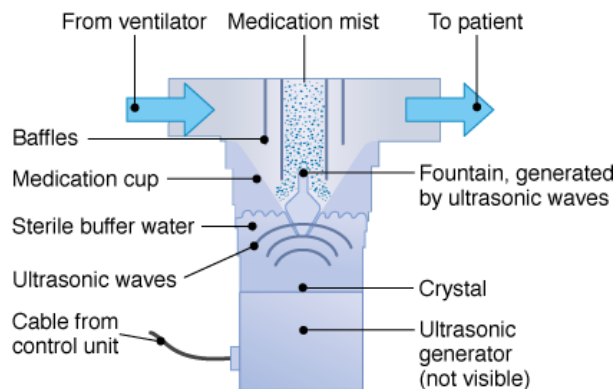
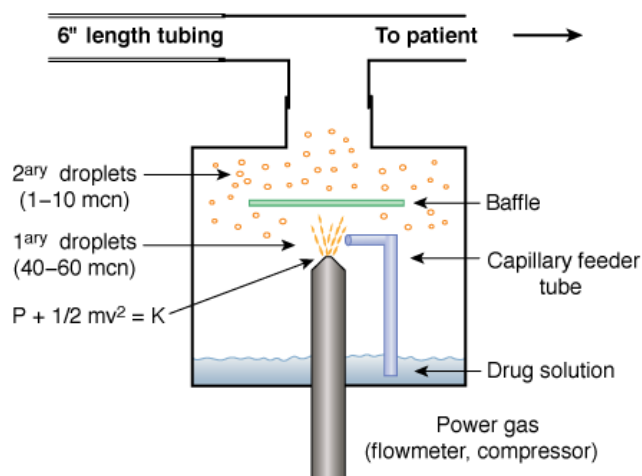
Factors influencing aerosol delivery in mechanically ventilated patients. *MDI*, metered-dose inhaler. (Modified with permission from Dhand R. Basic techniques for aerosol delivery during mechanical ventilation. *Respir Care*. 2004;49(6):611–622.)

Aerosol-Generating Devices

Nebulizers

Several types of nebulizers, jet, ultrasonic, vibrating mesh, and soft-mist inhalers convert liquids into aerosols for inhalation (Fig. 63-2).⁶¹

Figure 63-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Several types of nebulizers are available for clinical use, including jet nebulizers (*left*), ultrasonic nebulizers (*upper right*), and vibrating mesh nebulizers (*lower right*). The jet nebulizers employ compressed gas or air to generate an aerosol. In ultrasonic nebulizers, aerosol is generated by vibration of a piezoelectric crystal at ultrasonic frequencies. In the vibrating mesh nebulizers, aerosol is produced by high-frequency vibration of a plate with multiple apertures. Vibration of the aperture plate pushes the liquid to be nebulized through the apertures and generates a fine particle mist. *mcn*, micron.

Jet Nebulizers

Aerosol particle size is influenced by nebulizer design, solution characteristics (density, viscosity, and surface tension) and volume, gas pressure and flow, baffle design, and ratio of liquid to gas flow.¹⁴⁹⁻¹⁵¹ Droplet size decreases when gas flow increases, whereas droplet size increases with increase in the ratio of liquid to gas flow. A certain volume of solution (dead or residual volume) cannot be nebulized. Residual volume varies from 1 to 3 mL; it can be reduced by using a nebulizer with a conical shape, improving the wetness of the plastic surfaces, and reducing the internal surface area of the nebulizer.^{149,151} During operation, the solution concentration increases and its temperature decreases secondary to evaporative losses.^{152,153} Both increased solution concentration and cooling influence nebulizer output and particle size.^{151,154} Significant disadvantages of jet nebulizers are the requirement for a power source, inconveniently long treatment time, need for equipment setup and cleaning, and significant variations in the performance of various nebulizers, both within the same brand and across different brands.¹⁵⁵⁻¹⁵⁷

Ultrasonic Nebulizers

Most ultrasonic nebulizers have a higher rate of nebulization and require a shorter time of operation than jet nebulizers (see Fig. 63-2). Generally, the aerosol particle size is larger with ultrasonic nebulizers compared to jet nebulizers. The cost and bulk of ultrasonic nebulizers and their relative inefficiency in nebulizing drug suspensions are major limitations to their use, although smaller ultrasonic nebulizers are available and have been employed during mechanical ventilation.¹⁵⁸⁻¹⁶²

Vibrating Mesh Nebulizers

Newer-generation nebulizers employ a vibrating mesh or plate with multiple apertures to produce an aerosol.^{163,164} Because the frequency of vibration of the plates is lower than that in ultrasonic nebulizers, these devices can be operated with a battery pack and or electrical source. As a result, these devices are portable and less noisy. Moreover, these devices have negligible residual volume, and this property significantly improves the drug output. The Aeroneb Pro (Aerogen Inc., Mountain View, CA) is specifically designed as an inline nebulizer (see Fig. 63-2); a breath-synchronized version of the

Aeroneb Pro—the Pulmonary Drug Delivery System (PDDS, Nektar Therapeutics, San Francisco, CA)¹⁶⁵—has been successfully employed in ventilated patients.¹⁶⁶ The vibrating mesh nebulizers have a high rate of nebulization, and drug output is two to four times higher than with jet nebulizers.¹⁶⁷ Unlike ultrasonic nebulizers, the temperature of the solution does not change during operation of the vibrating mesh nebulizers, and proteins and peptides can be nebulized with minimal risk of denaturation.

Intratracheal Catheter

The intracorporeal nebulizing catheter (Aeroprobe; Trudell Medical International, London, Ontario, Canada) is a novel, investigational device that produces an aerosol in the trachea.^{168,169} Preliminary data suggest that lung deposition is improved with its use compared to more conventional forms of aerosol administration.^{170,171}

Pressurized Metered-Dose Inhalers

The pMDI canister contains a pressurized mixture of propellants, surfactants, preservatives, flavoring agents, and active drug, the latter comprising approximately 1% of the total contents.^{172,173} This mixture is released from the canister through a metering valve and stem, which fits into an actuator boot.^{172,173} Previously, most pMDIs used chlorofluorocarbon (CFC) propellants, but a newer generation of pMDIs contain hydrofluoroalkane (HFA) propellants.¹⁷⁴ The formulation, metering valve, and actuator design of HFA-pMDIs are different from those of CFC-pMDIs.⁶¹

The efficiency of drug delivery with a pMDI depends on how well the canister stem fits into the spacer adapter. In bench models of mechanical ventilation, HFA-pMDIs used with an AeroVent spacer provided lower drug delivery than that achieved with CFC-pMDIs.¹⁷⁵ In contrast, beclomethasone HFA-pMDIs employed with an AeroChamber HV MV spacer (Monaghan Medical, Plattsburgh, NY) had a higher efficiency of drug delivery than the beclomethasone CFC-pMDI.¹⁷⁶ Likewise, HFA-pMDIs were shown to achieve drug delivery similar to CFC-pMDIs in pediatric and neonatal bench models of mechanical ventilation.^{177,178} To improve drug delivery with HFA-pMDIs during mechanical ventilation, actuators that better fit the stem of the HFA-pMDI canister must be employed.

Dry Powder Inhalers

Dry powder inhalers could be employed inline in ventilator circuits either by employing the ventilator's inspiratory airflow to generate an aerosol or by first producing an aerosol and then entraining the drug particles into the airflow from the ventilator. Everard et al¹⁷⁹ employed a modified Turbuhaler in a dry ventilator circuit, and found that approximately 20% of the nominal dose was delivered to a filter placed at the distal end of the endotracheal tube. Humidity reduces drug delivery from dry powder inhalers,¹⁸⁰ and because ventilated patients routinely receive warm and humidified gas, the feasibility of administering dry powders during mechanical ventilation needs further evaluation.

Face Masks

Aerosol deposition in nasal passages significantly reduces drug delivery to the lung^{181–183} and could reduce bronchodilator efficacy;¹⁸⁴ however, face masks may be necessary for treatment of acutely dyspneic or uncooperative patients. For optimal efficacy, the face mask should produce a tight seal^{185–187} to avoid aerosol leakage and increased aerosol deposition around the eyes.¹⁸⁸

The orientation of the nebulizer with respect to the face mask influences the pattern of aerosol deposition. In “front-loaded” masks the nebulizer is inserted directly into the face mask, whereas in “bottom-loaded” masks the aerosol enters the mask from below. Front-loaded masks provide greater inhaled mass but also produce greater facial and ocular deposition.¹⁸⁹ Deposition of aerosol on the face and eyes could be minimized by employing masks that incorporate vents and have cut outs in the region of the eyes.^{189,190}

High-Flow Nasal Cannulae

Humidified high flow nasal cannulae are increasingly employed in the ICU to enhance gas exchange and avoid mechanical ventilation.¹⁹¹ In a bench study, the inhaled mass of aerosol and aerosol particle size with high-flow nasal cannulae were comparable to those obtained with mouthpiece inhalation from a continuously operating jet nebulizer.¹⁹²

Methods to Assess Aerosol Delivery during Mechanical Ventilation

In the past, pMDIs and nebulizers^{193,194} were shown to have poor efficiency during mechanical ventilation, mainly because of drug deposition in the ventilator circuit and artificial airway.^{195,196} Both in vitro and in vivo studies have helped in understanding the complex factors governing aerosol

delivery during mechanical ventilation.^{196–198}

Whereas in vitro methods measure *drug delivery* to the lower respiratory tract, in vivo methods measure the amount of *drug deposition* in the lung. This distinction is important because a variable portion of inhaled particles do not deposit in the lung and are exhaled. A “mass balance” technique that matches ventilator circuits and ventilator parameters has been employed to determine the correlation between the results of in vitro tests and those in ventilator-supported patients.^{199,200} With such techniques, it was estimated that approximately 5% of the nominal dose of albuterol administered by a pMDI is exhaled by ventilated patients,¹⁷⁵ compared to less than 1% exhaled by ambulatory patients.²⁰¹ The mean exhaled fraction (7%) with nebulizers in ventilated patients is similar to that with pMDIs, but there is considerable variability (coefficient of variation: 74%) among patients.¹⁹⁹

The particle size of the aerosol is an important determinant of aerosol delivery to the lung. Devices that produce aerosols with mass median aerodynamic diameter less than 2 μm are more efficient during mechanical ventilation than devices that produce aerosols with larger particles.^{199,200} Nebulizers that produce smaller particle size have been employed, but they require a considerably greater time to deliver a standard dose.^{200,202} Moreover, a significant proportion of submicronic particles (<1 μm) are exhaled.¹⁹⁹ For optimal pulmonary deposition, the size of the particles in the aerosol should be small enough to allow maximum penetration through the artificial airway, yet large enough to avoid being carried back out into the atmosphere with the exhaled breath.

In Vitro Studies

Carefully performed in vitro tests that simulate the conditions of actual clinical use have played an important role in determining the optimal techniques for administering aerosols to ventilated patients.^{175,176,199,200,202–209} Table 63-2 shows various bench models that were used. Models that employ a tracheobronchial model and directly measure the amount of drug deposited on a filter placed distal to the endotracheal tube^{175,176,195,209–211} have produced the most reproducible results.

Table 63-2: Determination of Lower Respiratory Tract Deposition of Aerosol Delivered by a Pressurized Metered-Dose Inhaler Using In Vitro Models

Type of model and Ref.	Type of Adapter	Breath Type	Measurement	Results
ETT (6.0, 7.5 and 9.0 mm) in trachea ²⁰⁴	Swivel adapter	Continuous flow or MDI actuation then flow	Filter weight	Greater efficiency with larger ETT and actuation into continuous flow
ETT and laser spectrometer ²⁰⁵	Three different adapters inline or cylindrical spacer	V_T 800 mL; flow 60 L min^{-1}	Particle volume 1 to 5 μm	Adapters produced lower volume of particles than standard actuator
Ventilator circuit; ETT (8 mm) ²⁰⁹	Swivel adapter at ETT or cylindrical spacer	V_T 800 mL; flow 48 L min^{-1}	Albuterol assay	Greater deposition with cylindrical spacer
ETT and laser spectrometer ²⁰⁶	Nine different MDI spacers or adapters	–	Particle volume 0.7 to 5.0 μm	Chamber spacers delivered greater volume than other adapters
Ventilator circuit ²⁰⁷	MDI with large chamber or small chamber spacer	V_T 700 mL; flow 50 L min^{-1}	Radioactivity	Similar delivery with the devices
Plastic syringe and simulated carina ²¹⁰	MDI with catheter	–	Albuterol assay	≈90% of dose delivered beyond ETT
ETT (6 mm) and swivel adapter ²¹¹	Catheters placed in ETT (13 or 22 cm long)	Flow 30 L min^{-1}	Albuterol assay	Longer catheters delivered greater dose than shorter catheters
Model of trachea and main bronchi ¹⁹⁵	Cylindrical spacer 8 mm ETT	Flow 40 L min^{-1}	Albuterol assay	Decreased deposition with humidification and CMV breaths

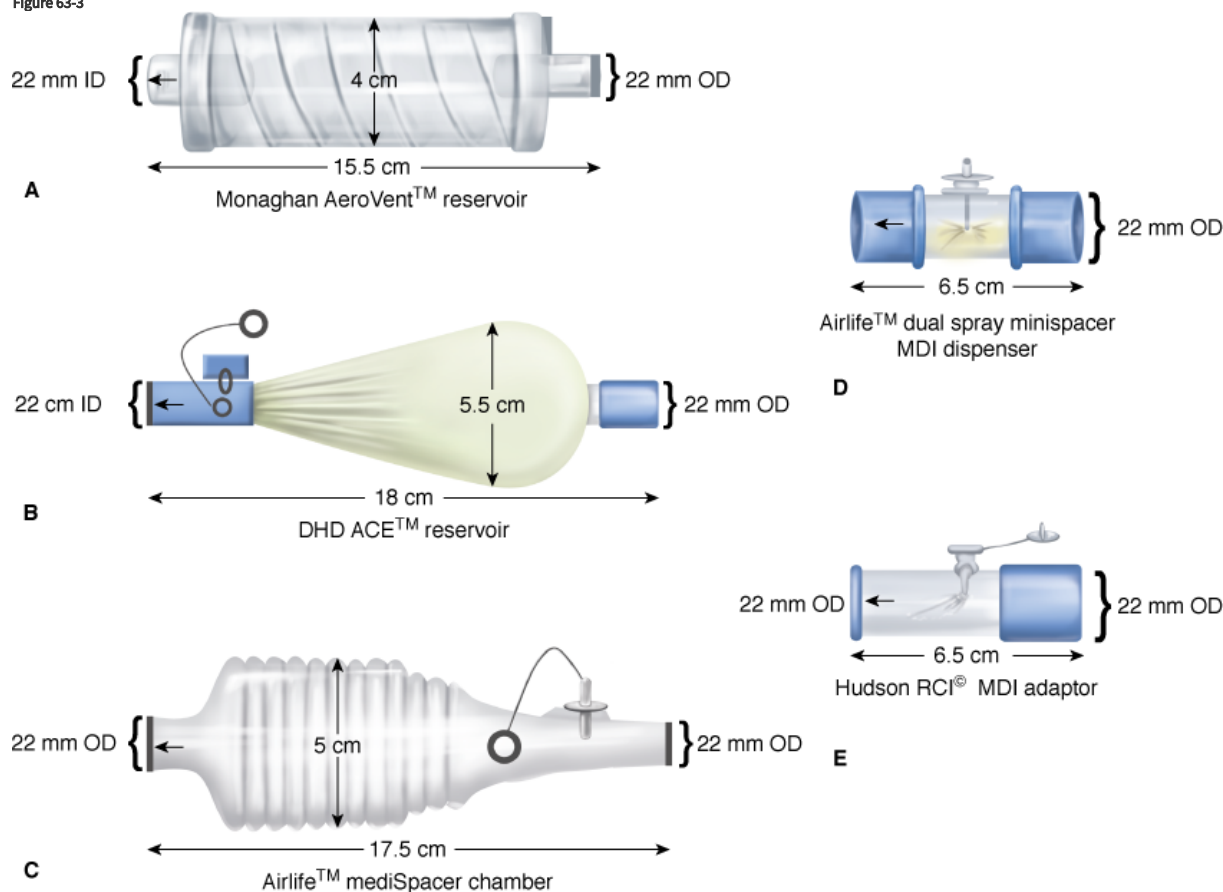
Abbreviations: CMV, controlled mechanical ventilation; ETT, endotracheal tube; pMDI, pressurized metered dose inhaler; V_T , tidal volume.

For any given aerosol-generating device, the efficiency of drug delivered varies widely; for pMDIs, it varies from 0.3% to 97.5% and for nebulizers from 0% to 42%. These variations in drug delivery underscore the need for optimizing the techniques of administration with each device. With pMDIs, the type of pMDI propellant formulation¹⁷⁶ and the drug formulation²⁰³ also influence drug delivery.

Configuration of the Device with Metered-Dose Inhalers

For a pMDI to be employed in a ventilator circuit, the canister must be removed from the actuator (supplied by the manufacturer) and connected to the ventilator circuit with another different adapter, thereby making it a unique device with different aerosol characteristics and performance. Several types of adapters, including elbow adapters, inline devices that may be unidirectional or bidirectional, and chamber or reservoir adapters, are commercially available (Fig. 63-3).^{162,212} The adapter efficiency could have a significant influence on the dose required to produce a therapeutic effect.²¹³ Several investigators have shown that employing a chamber spacer with a pMDI in a ventilator circuit results in fourfold to sixfold greater aerosol drug delivery compared with either an elbow adapter or a unidirectional inline spacer.^{202,205,209,214} A pMDI and chamber spacer placed at a distance of approximately 15 cm from the endotracheal tube provides efficient aerosol delivery and elicits a significant bronchodilator response.^{12,197,215} The efficiency of a bidirectional inline spacer was higher than that of a unidirectional inline spacer²⁰³ and was comparable to that achieved with chamber spacers,²⁰³ although the performance of the bidirectional spacer has not been established in clinical studies.

Figure 63-3



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Commercially available spacers and adapters that are used to connect a metered-dose inhaler (MDI) canister in the ventilator circuit. *Top left*: Collapsible spacer chamber. *Middle left*: Aerosol cloud enhancer, wherein the aerosol plume is directed away from the patient. *Bottom left*: Noncollapsible spacer chamber. *Top right*: Bidirectional actuator (mini spacer). *Bottom right*: Inline adapter. OD, outside diameter. (Used, with permission, from Rau et al.²⁰³)

Configuration of the Device with Nebulizers

Both jet and ultrasonic nebulizers are connected in the inspiratory limb of the ventilator circuit or at the patient Y. Placing a jet nebulizer at a distance from the endotracheal tube improves its efficiency compared with placing it between the patient Y and endotracheal tube.^{208,216,217,218} Placement of the continuously operating jet nebulizer before the humidifier (i.e., between the humidifier and ventilator) had a higher efficiency of aerosol delivery than placement closer to the endotracheal tube,²¹⁸ probably because the inspiratory limb of the ventilator circuit acts as a reservoir for the aerosol during the exhalation phase. For the same reason, addition of a reservoir between the nebulizer and endotracheal tube also modestly increases efficiency of drug

delivery.²¹⁹ The nebulizer brand,^{157,208} diluent volume, operating pressures and flows, and duration of treatment,^{149,157} influence the efficiency of aerosol generation.

Ultrasonic nebulizers are infrequently employed for bronchodilator therapy during mechanical ventilation and there is scant published information about drug delivery with these devices.¹⁶⁰ Moreover, the particle size of aerosols produced by ultrasonic nebulizers in ventilator circuits has not been well characterized. The newer vibrating mesh nebulizers deliver twofold to fourfold higher drug dose than jet nebulizers. In the absence of bias flow, a vibrating mesh nebulizer was most efficient for drug delivery when it was placed 15 cm from the endotracheal tube, whereas in the presence of bias flow (2 L/min) the efficiency of the device was enhanced by placing it at the inlet of the humidifier (i.e., between the humidifier and ventilator).²¹⁸

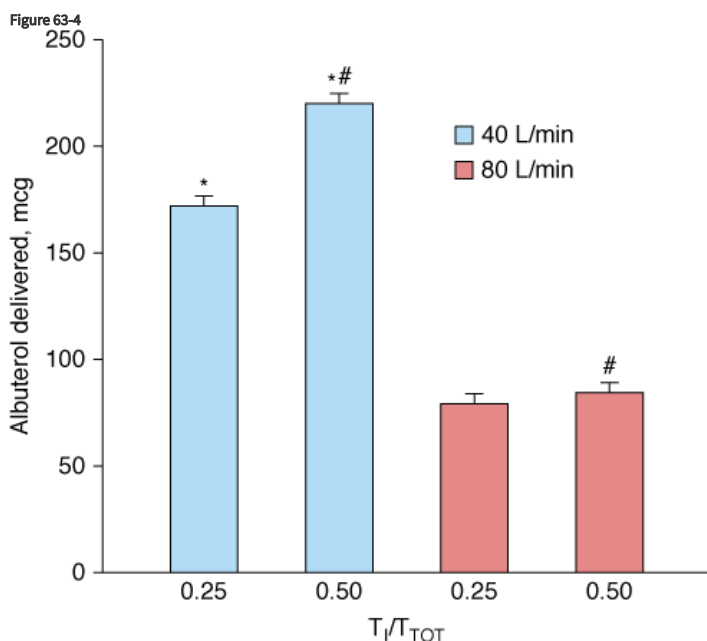
Synchronizing Aerosol Generation with Inspiratory Airflow

The actuation of a pMDI must be synchronized with the precise onset of inspiratory airflow from the ventilator.^{220,221} As short as a 1- to 1.5-second delay between pMDI actuation and a ventilator breath can profoundly reduce the efficiency of drug delivery.²⁰²

In a ventilator circuit, nebulizers can be operated continuously or intermittently by airflow from the ventilator. Continuous aerosol generation requires a pressurized source of gas (from a wall outlet, pressurized tank, or an air compressor), whereas intermittent operation that is synchronized with inspiratory airflow requires a separate line to conduct inspiratory airflow from the ventilator to the nebulizer. Intermittent operation of the nebulizer is more efficient for aerosol delivery compared with continuous aerosol generation because it minimizes aerosol wastage during the exhalation phase of the breathing cycle.^{200,217} The lower driving pressure provided by the ventilator (<15 pounds per square inch [psi]) than that provided by pressurized gas (≥ 50 psi) could decrease the efficiency of some nebulizers.²²² Aerosol generated by a nebulizer operating at the lower pressure may generate particles whose diameter is larger than the 1 to 5 μm that is optimal for aerosol deposition. When intermittent nebulizer operation is employed, the specific ventilator and nebulizer brand should be tested to determine the characteristics of the aerosol generated and the efficiency of drug delivery.²⁰⁰

Ventilator-Related Factors

The characteristics of the ventilator breath have an important influence on aerosol drug delivery. A tidal volume of 500 mL or more (in an adult),¹⁹⁵ longer inspiratory time, and slower inspiratory flows improve aerosol delivery^{195,223,224} (Fig. 63-4). An inspiratory flow rate of 30 to 50 L/min is optimal for aerosol delivery; however, slow inspiratory flow rates may increase inspiratory time and by reducing the time for exhalation may have the unintended consequence of increasing auto-positive end-expiratory pressure.²²⁵ Drug delivery is linearly correlated with a longer duty cycle (inspiratory time-to-total time [T_I/T_{TOT}]) for both pMDIs and nebulizers.^{175,195,208} Moreover, drug delivery is improved when a pMDI is synchronized with a simulated spontaneous breath compared with a controlled ventilator breath of similar tidal volume.

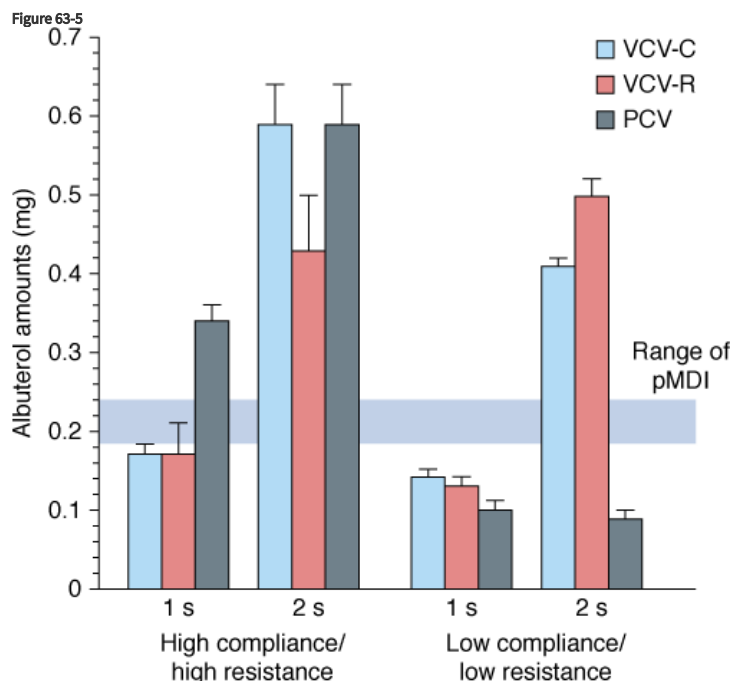


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Comparison of aerosol delivery at different inspiratory airflows and duty cycles (T_I/T_{TOT}) in a bench model of mechanical ventilation. The ventilator delivered a tidal volume of 1000 mL with a constant inspiratory flow of 40 or 80 L/min, and the frequency of breathing was varied to achieve T_I/T_{TOT}

values of 0.25 or 0.50 at each inspiratory flow setting. Albuterol delivery to the bronchi was greater with a T_I/T_{TOT} of 0.50 than of 0.25 at inspiratory flows of 40 L/min and 80 L/min. For each value of T_I/T_{TOT} , drug delivery with a slower inspiratory airflow (40 L/min) was almost twice that at the faster inspiratory airflow (80 L/min). $^*, p < 0.01$, 40 L/min versus 80 L/min at T_I/T_{TOT} of 0.25 and T_I/T_{TOT} of 0.50; $\#, p < 0.01$ T_I/T_{TOT} of 0.5 versus T_I/T_{TOT} of 0.25 at 40 L/min and 80 L/min. (Dhand R. Aerosol delivery during mechanical ventilation: from basic techniques to new devices. *J Aerosol Med Pulm Drug Deliv.* 2008;21:45–60.)

The inspiratory waveform influences drug delivery from nebulizers, but has much less influence on drug delivery from a pMDI.²²⁶ Unlike pMDIs, nebulizer efficiency could be different during pressure-controlled ventilation than during volume-controlled ventilation (Fig. 63-5).



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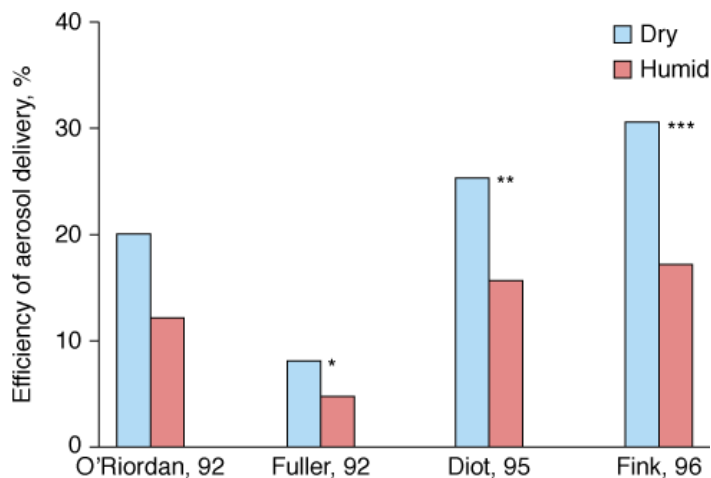
Comparison of aerosol delivery from a pressurized metered-dose inhaler (pMDI) and jet nebulizer in bench models of pressure-controlled and volume-controlled ventilation. The lung mechanics were varied by selecting two settings of resistance and compliance to achieve high or low time constants. For each condition, the amounts of aerosol delivered during inspiratory times of 1 second or 2 seconds were measured. Increasing the duration of inspiration from 1 second to 2 seconds improved the nebulizer efficiency. In the high-compliance/high-resistance setting with 1-second inspiration, nebulizer efficiency was higher during pressure-controlled than during volume-controlled ventilation, whereas the converse occurred in the low-compliance/low-resistance setting with 2 seconds inspiratory time. In contrast, the efficiency of a pressurized MDI (horizontal purple area) remained fairly constant under the various conditions simulated in the bench model. Thus, several factors, such as inspiratory time, pattern of inspiratory flow, and lung mechanics, that could influence drug delivery from a nebulizer have minimal influence on drug delivery from a pMDI. PCV, pressure-controlled ventilation; VCV-C, volume-controlled ventilation with a constant inspiratory flow; VCV-R, volume-controlled ventilation with a descending ramp flow pattern. (With kind permission from Springer Science and Business Media: Hess DR, Dillman C, Kacmarek RM. In vitro evaluation of aerosol bronchodilator delivery during mechanical ventilation: pressure-control vs. volume control ventilation. *Intensive Care Med.* 2003;29:1145–1150.)

The breath-triggering mechanism does not significantly influence drug delivery from a pMDI, but use of a flow trigger with a nebulizer could dilute the aerosol and increase the washout of the aerosol into the expiratory limb between breaths.^{195,208}

Circuit-Related Factors

Several investigators found that drug delivery to the lower respiratory tract from both pMDIs and nebulizers is reduced by 40% or more in a humidified circuit when compared to a dry circuit (Fig. 63-6).^{158,175,200,202,207,208} Circuit humidity increases the size of drug particles generated by a nebulizer.²²⁷ When a pMDI is employed in a ventilator circuit, humidity probably interferes with propellant evaporation so that drug particles remain of a larger size and impaction losses are increased.^{212,228}

Figure 63-6



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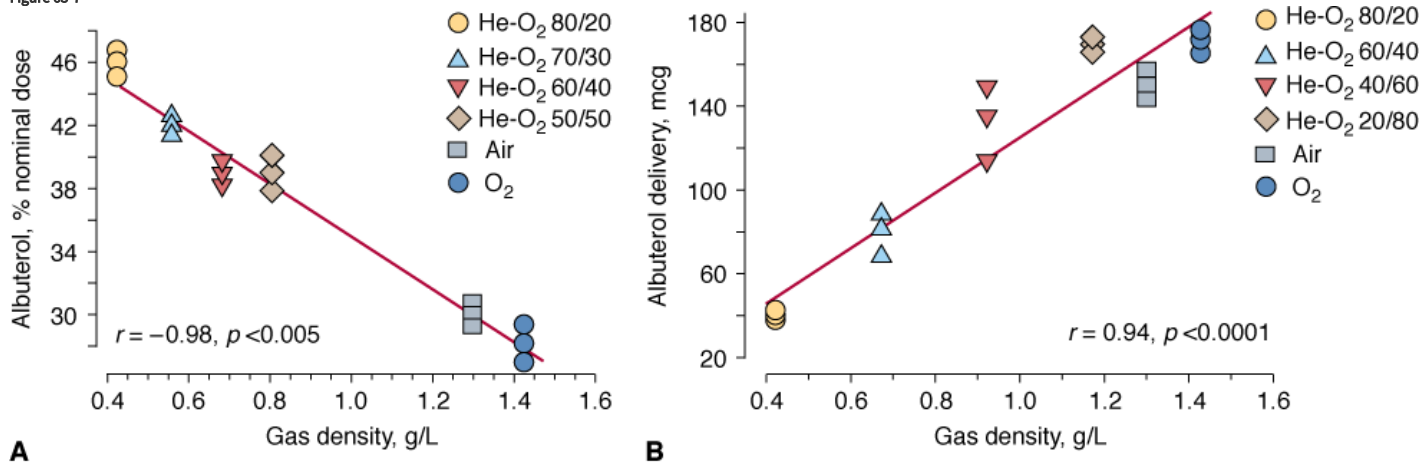
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Effect of humidity on aerosol delivery. The delivery of aerosol to the lower respiratory tract in bench models of mechanical ventilation is reduced by approximately 40% when the circuit is humidified instead of dry. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. Studies: O'Riordan et al. 1994, Fuller et al. 1992, Diot et al. 1995, and Fink et al. 1996, (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Dhand R, Tobin MJ. Inhaled bronchodilator therapy in mechanically ventilated patients. *Am J Respir Crit Care Med*. 1997;156:3–10. Official Journal of the American Thoracic Society.)

Heat and moisture exchangers (HMEs) are employed as an alternative to heated humidifiers.²²⁹ The filter in the HME captures the heat and moisture in the exhaled breath and transfers part of it to the air in the following inspiration. The HME filter captures drug particles in the aerosol and markedly reduces the efficiency of drug delivery. Therefore, the HME must be removed from the circuit during aerosol treatments. Although placement of a nebulizer between the HME and endotracheal tube could provide adequate drug delivery, backflow of aerosol from the nebulizer could deposit on the HME filter, increase its airflow resistance, and increase the work of breathing for the patient.²³⁰

The density of the inhaled gas also influences drug delivery. High inspiratory flows employed during mechanical ventilation are associated with turbulence. Inhalation of a less-dense gas, such as a helium-oxygen 70/30 mixture, makes airflow less turbulent and more laminar. The use of helium-oxygen mixtures improved aerosol drug delivery in a pediatric model of mechanical ventilation.²³¹ In a bench model of adult mechanical ventilation, drug delivery from a pMDI was noted to be 50% higher with a helium-oxygen 80/20 mixture than with oxygen alone (Fig. 63-7A).²³² In contrast, nebulizer operation with helium-oxygen reduced drug output and respirable mass (Fig. 63-7B).^{232,233} A practical method to achieve maximum pulmonary deposition of aerosol from a nebulizer during mechanical ventilation is to operate the nebulizer with oxygen at a flow rate of 6 to 8 L/min and to entrain the aerosol generated into a ventilator circuit containing helium-oxygen (Fig. 63-8).²³³ Before employing helium-oxygen in ventilated patients, it is important to ensure that the ventilator is compatible with the use of such gas mixtures.²³⁴

Figure 63-7



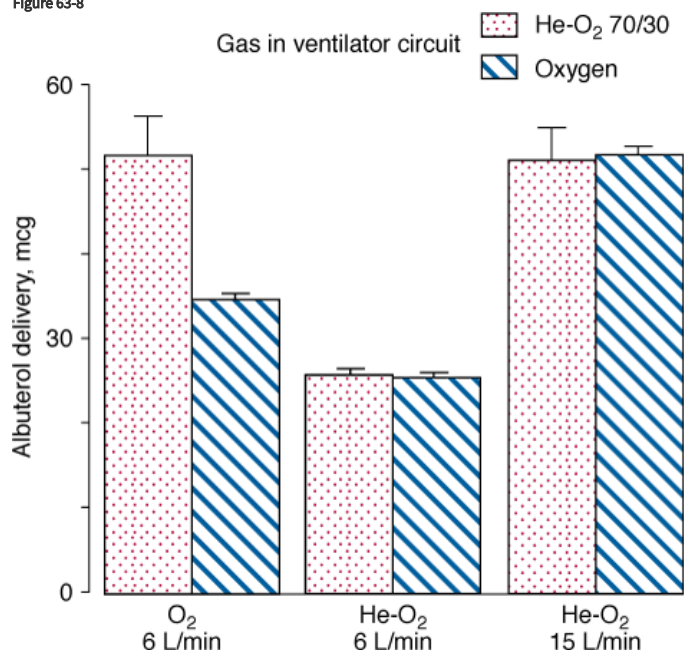
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Effect of gas density on aerosol delivery from a pressurized metered-dose inhaler (pMDI) and jet nebulizer. A. Albuterol was administered via a pMDI and chamber spacer in an unheated dry ventilator circuit containing air, 100% oxygen, or several mixtures of helium-oxygen (80/20, 70/30, 60/40, and 50/50). Albuterol delivery from a pMDI (percentage of nominal dose) was inversely related ($r = -0.98$; $p < 0.005$) to gas density. B. Albuterol was administered with

a jet nebulizer operated at a constant flow of 6 L/min of air, 100% oxygen, or helium-oxygen mixtures (as above) and albuterol output from the nebulizer was measured. Albuterol output from the nebulizer was positively correlated ($r = 0.94$; $p < 0.0001$) with the density of gas used to operate the nebulizer. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Goode ML, Fink JB, Dhand R, et al. Improvement in aerosol delivery with helium-oxygen mixtures during mechanical ventilation. *Am J Respir Crit Care Med*. 2001;163:109–114. Official Journal of the American Thoracic Society.)

Figure 63-8



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Effect of gas density and operating flow on aerosol delivery from a nebulizer. A jet nebulizer was operated with 100% oxygen at 6 L/min, helium-oxygen (70/30) at 6 L/min, or helium-oxygen (70/30) at 15 L/min. Drug delivery on filters was measured with the circuit containing helium-oxygen (stippled bars) or oxygen (hatched bars). Albuterol delivery was greatest when the nebulizer was operated with oxygen and the ventilator circuit contained helium-oxygen. Bars represent standard error (SE). (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Goode ML, Fink JB, Dhand R, et al. Improvement in aerosol delivery with helium-oxygen mixtures during mechanical ventilation. *Am J Respir Crit Care Med*. 2001;163:109–114. Official Journal of the American Thoracic Society.)

Aerosol impaction on the endotracheal tube poses a significant barrier to effective drug delivery in infant and pediatric mechanical ventilation (endotracheal tube internal diameter 3 to 6 mm),^{204,235} with aerosol delivery efficiency being lower with narrower endotracheal tube tubes.²³⁶ In adult mechanical ventilation, there was no difference in nebulizer efficiency with endotracheal tubes of internal diameter 7 mm versus internal diameter 9 mm.²⁰⁸ Drug losses within the endotracheal tube could be minimized by placing the aerosol generator at a distance from the endotracheal tube instead of being directly connected to it.¹⁹⁶ Removing the right angled elbow connector between the patient Y and endotracheal tube improves aerosol delivery.^{175,237} Some investigators attached a long catheter to the nozzle of an pMDI and delivered aerosol directly into the trachea (i.e., beyond the endotracheal tube).²¹¹ With this delivery system, concerns have been raised about catheter blockage and mucosal damage induced by propellants, surfactant, or other constituents of the pMDI formulation.²³⁸

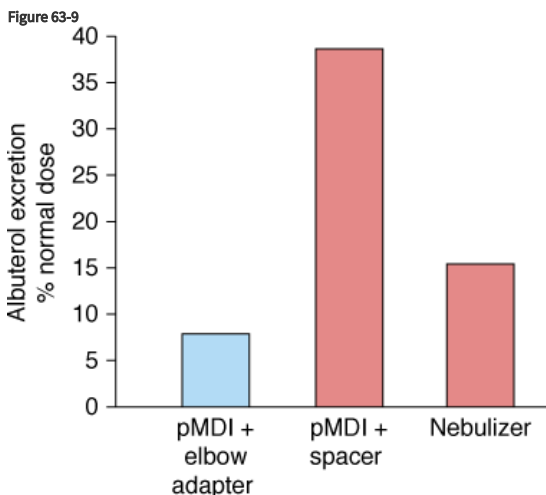
In Vivo Studies

Several investigators have used radionuclides and measured plasma or urinary drug levels to determine pulmonary deposition of aerosols in ventilated patients.

Radionuclide Studies

In ventilated patients, the pulmonary deposition of nebulized radiolabeled aerosol has been variously reported to be $1.22 \pm 0.4\%$,¹⁹⁴ $2.22 \pm 0.8\%$,¹⁵⁸ $2.9 \pm 0.7\%$,¹⁹³ and $15.3 \pm 9.5\%$.¹⁹⁹ Several factors, including type of radiolabel used, nebulizer brand, treatment time, circuit humidity, and methods used to calculate the amount of aerosol deposition,^{199,200,208} contribute to the reported variation. With a pMDI and spacer chamber, approximately 6% of the dose was deposited in the lower respiratory tract;^{194,214} this value was significantly lower than reported values (10% to 20%) with a pMDI and spacer in nonintubated ambulatory patients.^{239,240}

Unlike nonintubated patients, direct deposition of aerosol in the oropharynx and subsequent enteral absorption cannot occur in ventilated patients. Therefore, estimation of plasma levels of drugs administered by a pMDI should reflect lower respiratory tract deposition. After administration of albuterol with a pMDI and spacer, the area under the concentration-time curve was marginally lower in the patients than in healthy controls.²⁴¹ Moreover, measurement of urinary albuterol excretion in thirty ventilated patients with normal renal function showed that albuterol recovery was highest (38%) after administration with a pMDI and chamber spacer, intermediate with a nebulizer (16%), and lowest (9%) with a pMDI and right-angle port connected to the endotracheal tube (Fig. 63-9).²⁴² These results corroborated previous investigations that found very low efficiency of drug delivery with the adapter connected to the endotracheal tube.^{208,209,214} Thus, pharmacokinetic studies show that with an optimal technique of administration, pulmonary drug deposition in ventilated patients is comparable to that achieved in healthy controls.



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Comparison of systemic bioavailability of albuterol administered by pressurized metered-dose inhaler (pMDI) and right-angle elbow adapter, pMDI and chamber spacer, or jet nebulizer. Urine was collected for 6 hours after drug administration and the amounts of albuterol and its sulfate conjugate determined. The efficiency of the delivery device was determined by determining the percentage of drug excreted. The three delivery systems varied in the efficiency of drug delivery. The pMDI and elbow adapter had the lowest efficiency, the jet nebulizer was intermediate, and the pMDI with chamber spacer had the highest efficiency. (Data from Marik et al.²⁴²)

Pharmacokinetic studies have also been employed to compare the efficiency of drug delivery during mechanical ventilation. Moraine et al measured urinary *ipratropium* levels after ultrasonic nebulization in ventilated patients, and found no differences in urinary *ipratropium* levels with the nebulizer placed close to the patient Y or when it was placed before the heated humidifier.²⁴³

Reconciling In Vitro Estimates of Drug Delivery with In Vivo Estimates of Drug Deposition

The wide variation between in vitro^{195,207-209} and in vivo estimates¹⁹⁴ of device efficiency can be reconciled by taking into account the effects of humidity in the ventilator circuit,¹⁹⁵ the quantity of exhaled aerosol that is not included in the in vitro measurement,¹⁷⁵ and by correcting for the quenching of radioactivity by the tissues of the chest wall.²¹⁴ When these factors are accounted for, in vitro data obtained with humidified ventilator circuits and in vivo gamma scintigraphic studies reveal that approximately 11% of the nominal dose from a pMDI and spacer chamber is deposited in the lower respiratory tract of ventilated patients. This value is remarkably close to values observed with the optimal use of a pMDI without a spacer (10% to 14%) in ambulatory patients.^{239,240}

Drug delivery from nebulizers also shows discrepancies between values obtained with bench models versus those obtained by gamma scintigraphy.^{158,193,194,219} Miller et al²⁰⁰ found that accounting for circuit humidity and breath-actuated nebulization could reconcile most observed differences.

Thus, in vitro investigations that accurately simulate the clinical settings could play an important role in determining the performance of various inhalation devices under a variety of conditions encountered during mechanical ventilation.

Clinical Use of Bronchodilators

Bronchodilators Used

Bronchodilators are among the most commonly used drugs in the ICU.¹³ In ventilator-supported patients with asthma or COPD, the goals of bronchodilator therapy are to reverse bronchoconstriction, decrease work of breathing, and/or relieve dyspnea. A response has been observed after administration of either aerosolized β -adrenergic^{1,9,10,12,21,129,213,215,244–255} or anticholinergic bronchodilators.^{129,252,256,257} Inhaled isoproterenol,^{244,258} isoetharine,²⁵⁹ metaproterenol,² fenoterol,^{245,252} albuterol,^{9,10,12,21,213,215,246–251,253–255} and salmeterol²⁹ produce significant bronchodilation in ventilated patients. The combination of fenoterol and ipratropium bromide was found more effective than ipratropium in ventilated patients with COPD.²⁶⁰

Selection of Patients

Ventilated patients with COPD demonstrate a significant decrease in airway resistance after administration of bronchodilators.^{12,21,129,215,245,248,250,251,253,254,256,257} Bronchodilators have been successfully used to treat acute bronchospasm in the operating room,^{244,258,259} and they are widely used in ventilated patients with severe asthma.²⁶¹ In addition, expiratory flow improved after bronchodilator administration in a heterogeneous group of ventilated patients.² Nebulized metaproterenol reduced elevated levels of airway resistance in patients with acute respiratory distress syndrome.^{4,5} Thus, a wide spectrum of patients receiving mechanical ventilation receive bronchodilators.³

Bronchodilator Efficacy

Assessment of Bronchodilator Response

The response to bronchodilators depends on several variables: patient airway geometry, degree of airway responsiveness, severity of disease, quantity of secretions, and counterregulatory effects of airway inflammation and other drugs. Most investigators assess response by measuring inspiratory airway resistance. Airway resistance in ventilated patients is commonly measured by performing rapid airway occlusions at constant flow inflation.^{262,263} This technique involves performing a breathhold at end-inspiration by occluding the expiratory port. Total or maximal inspiratory resistance (Rrs max) can be partitioned into minimal inspiratory resistance (Rrs min), which reflects “ohmic” resistance of the airways, and additional effective resistance (Δ Rrs). The latter resistance represents time-constant inhomogeneities within the lung (“pendelluft”) and viscoelastic behavior of the pulmonary tissues.^{262,263} Similarly, airway occlusion at end-expiration produces an increase in airway pressure to a plateau value, signifying intrinsic positive end-expiratory pressure.¹² Comparisons of airway resistance and intrinsic positive end-expiratory pressure before and after drug administration are useful for assessing response.

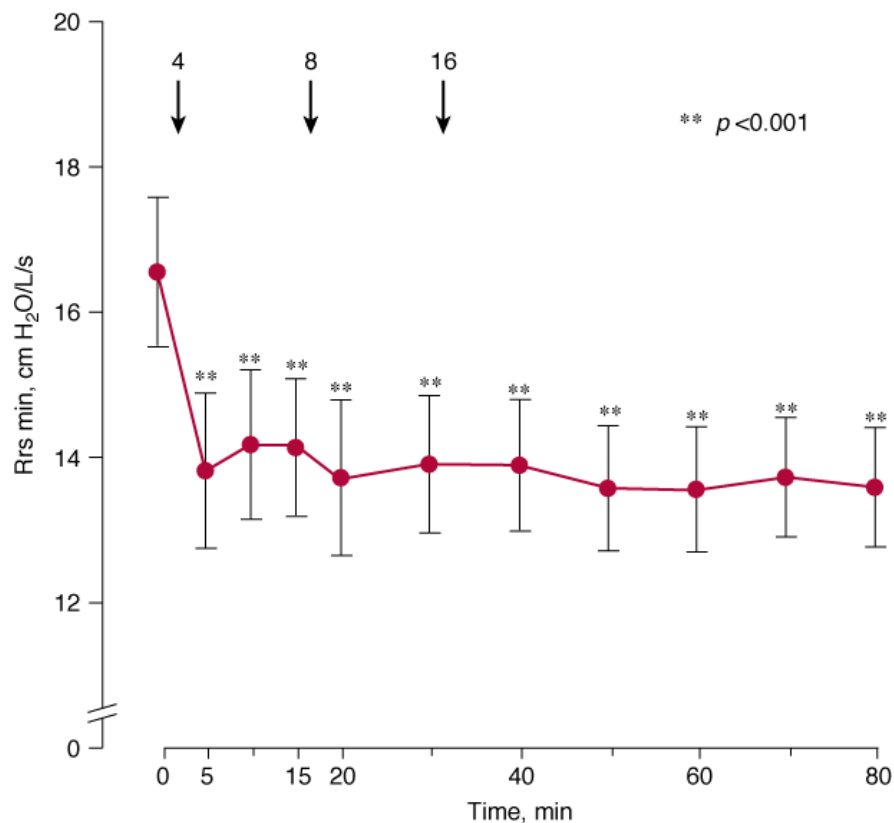
Most ventilated patients with COPD demonstrate a decrease in airway resistance and intrinsic positive end-expiratory pressure following bronchodilator administration.^{4,9,12,21,213,215,245,248,250,251–255} That Δ Rrs does not decrease significantly after albuterol delivery with a pMDI^{12,212,215} suggests that the bronchodilator effect occurs predominantly in the central airways with little effect on viscoelastic behavior or time-constant inhomogeneities in the lung. Moreover, albuterol does not significantly influence the elastic properties of the lung.¹² In contrast, a greater decline in Δ Rrs was noted after nebulizer delivery of albuterol and ipratropium bromide.²⁵² This difference in response between pMDIs and nebulizers could be due to higher drug deposition in peripheral airways with the use of a nebulizer.

In bench models, drug delivery was increased during simulated spontaneous breaths compared to controlled mechanical breaths.¹⁹⁵ In ventilated patients with acute exacerbations of COPD, however, the bronchodilator response did not differ between controlled mechanical ventilation and pressure support.²⁵⁵ The use of propofol for sedation was a significant confounder of that study because of its known bronchodilator effect.²⁶⁴

Dose Response to Bronchodilator Administration in Ventilated Patients

Few investigators have examined the dose response to bronchodilators in ventilated patients.^{213,215,224,245,254} Most investigators found that the response with the higher doses was no greater than that observed after the initial dose (Figs. 63-10 and 63-11).^{215,245} Dose–response curves in stable, ventilated patients with COPD are shallow; when the technique of administration is carefully executed, most patients achieved near maximal bronchodilation with 4 puffs of albuterol.²¹⁵ Mouloudi et al^{21,250,251,253} showed that, with an optimal technique of administration, alterations in mechanical ventilator settings do not influence the response to albuterol in stable patients with COPD. Patients with acute exacerbations of asthma or COPD may require higher doses.

Figure 63-10

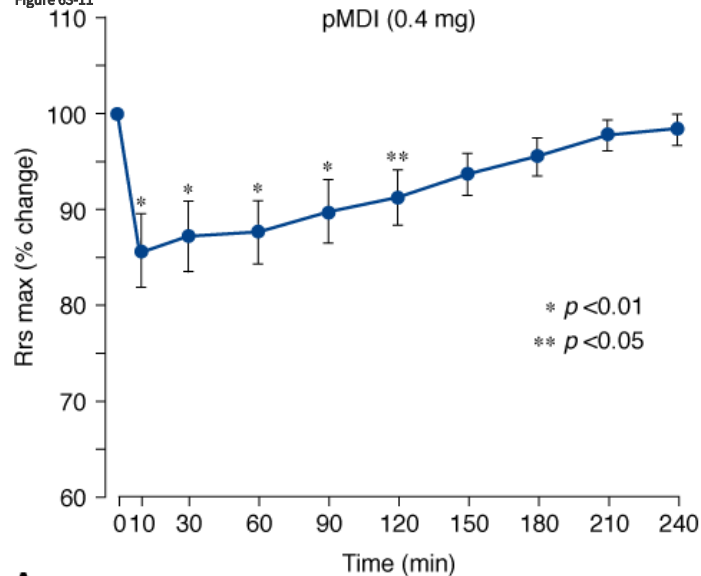


Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

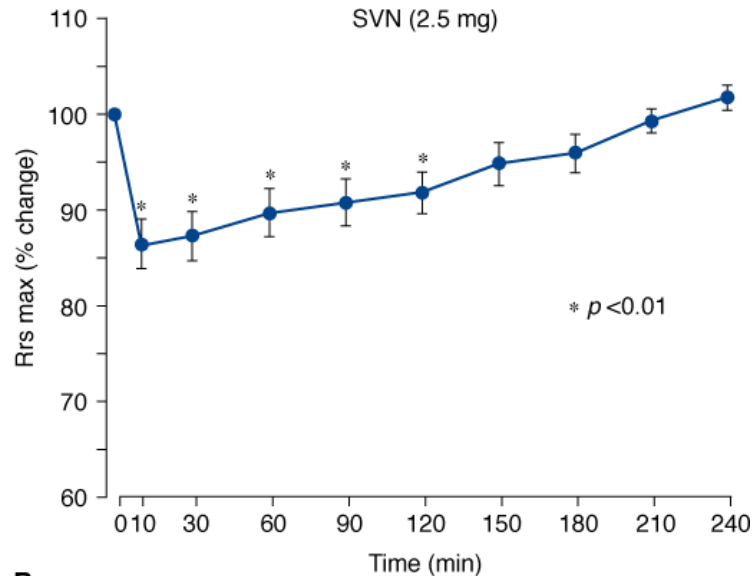
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Effect of albuterol on minimal inspiratory resistance (*Rrs min*) in twelve stable mechanically ventilated patients with COPD. Significant decreases in *Rrs min* occurred within 5 minutes of administration of 4 puffs of albuterol. The addition of 8 and 16 puffs (cumulative doses of 12 and 28 puffs, respectively) did not achieve a significantly greater effect than that with 4 puffs ($p > 0.05$). Bars represent standard error (SE). **, $p < 0.001$. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Dhand R, Duarte AG, Jubran A, et al. Dose-response to bronchodilator delivered by metered-dose inhaler in ventilator-supported patients. *Am J Respir Crit Care Med*. 1996;154:388–393. Official Journal of the American Thoracic Society.)

Figure 63-11



A



B

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Effect of albuterol on maximum inspiratory airway resistance (*Rrs max*) in stable, mechanically ventilated patients with COPD. There was a decrease in *Rrs max* from baseline values within 10 minutes of albuterol administration. **A.** Change in *Rrs max* from baseline (time 0) after 4 doses of albuterol from a pMDI. **B.** Change in *Rrs max* from baseline (time 0) after 2.5 mg of albuterol given by nebulizer. Significant reductions in *Rrs max* were sustained for 2

hours and returned to baseline by 4 hours. The response to albuterol administered by pMDI (0.4 mg) was comparable to that achieved with 2.5 mg administered by nebulizer. Bars represent standard error of mean (SEM). (Used, with permission, from Duarte et al.²⁵⁴)

Duration of Bronchodilator Response

In stable, ventilated patients with COPD, the bronchodilator effect of albuterol is sustained for 2 to 3 hours.^{21,254} Thus, in contrast to the four-times-a-day and as-needed albuterol dosing in ambulatory patients, ventilated patients may require dosing every 3 to 4 hours. Two or three times daily dosing may be feasible with salmeterol because its bronchodilator effect is sustained for approximately 8 hours in ventilated patients.²⁵⁵

Aerosol Therapy in Mechanically Ventilated Neonates and Infants

Infants breathe with a high frequency and low tidal volume. This rapid shallow breathing pattern reduces drug deposition in the lung by limiting the time available for particle deposition. In nonintubated, spontaneously breathing infants, pulmonary deposition of radiolabeled aerosols administered via pMDI and face mask, or nebulizer and face mask, is as low as 0.3% to 2.6% of the nominal dose.^{265–269} Several investigators report similar decreases in aerosol delivery in ventilated neonates and infants because of small-diameter endotracheal tubes and ventilator tubing, and the low tidal volumes employed.^{235,270,271} Moreover, ventilator modes differ; pressure-limited and time-cycled modes of ventilation are more common, with continuous flow through the circuit and very-short-duty cycles.

In Vitro Studies of Aerosol Drug Delivery in Infants

Metered-Dose Inhalers

When the pMDI was actuated into a prototype spacer chamber, there was a 10-fold variation (from 0.2% to 2.0% of the nominal dose) in drug delivery to a filter placed beyond the endotracheal tube.²⁷² Higher values of drug delivery were achieved when the pMDI was actuated before inspiration than after inspiration, and when higher tidal volumes and longer inspiratory times (inspiratory-to-expiratory timing [I:E] ratio 1:1) were employed. The type of spacer employed to actuate a pMDI has a dramatic influence on drug delivery (see Table 63-3).^{273–276} In a dry infant ventilator circuit, drug delivery as high as 14.2% was achieved when an AeroChamber MV15 (Monaghan Medical Corp., Plattsburgh, NY) was placed between the ventilator Y and endotracheal tube and the pMDI was actuated at end-expiration.²⁷³ With an ACE spacer in a humidified circuit during infant mechanical ventilation, an HFA-albuterol pMDI formulation gave marginally higher drug delivery than a CFC-albuterol pMDI formulation (5.7% vs. 4.8%, respectively).²⁷⁶ When a HFA-pMDI was employed with a valved-holding chamber made of electrostatic charge dissipative polymer, aerosol delivery was lower during simulated mechanical ventilation compared to delivery with a manual resuscitation bag or during simulated spontaneous breathing.²⁷⁷

Table 63-3: Efficiency of Drug Delivery with a pMDI and Spacer in Infant Ventilation

First author ^a	Spacer	ET(mm)	V _T (mL)	Mode	Efficiency (%)
Everard	Prototype	3.0	11 to 22	Pressure limited, 30 cm H ₂ O	1.5 to 2
Coleman	ACE	3.5	55	Pressure 60/5 cm H ₂ O	14.5
	AeroChamber MV				11.9
	AeroVent				6.8
	Inline adapter				6.4
Avent	AeroChamber	3.5	–	Pressure 20/2 cm H ₂ O	2.2
	Inline adapter			Humidified circuit	0.1
Lugo	CFC-pMDI with ACE	3.0	7	Pressure 25/4 cm H ₂ O	4.8
	HFA-pMDI with ACE			Humidified circuit	5.7
DiBlasi	HFA-pMDI with	2.5, 3.5, 4.0	6 to 60	Mechanical/manual	3.7 to 13.4 ^a
	AeroChamber Mini/Spontaneous breaths			Humidified circuit	

Abbreviations: ET, internal diameter of endotracheal tube; V_T, tidal volume

^a Values are means of efficiency of drug delivery in various settings. Efficiency of drug delivery was lower during mechanical ventilation compared to manual resuscitation or spontaneous breathing probably because of the effects of high humidity in the circuit during mechanical ventilation.

Source: Data from Everard,²⁷² Coleman,²⁷⁴ Avent,²⁷⁵ Lugo,²⁷⁶ and DiBlasi.²⁷⁷

Nebulizers

Nebulizers have shown uniformly poor efficiency for drug delivery in infants. During simulated infant ventilation, nebulizers deliver less than 2% of the nominal drug dose.^{273,274,278–280} Many factors influence drug delivery with a nebulizer: type of nebulizer^{273,278} and its position in the circuit,²⁷⁶ intermittent versus continuous nebulization,²⁸⁰ nebulizer flow,²⁷⁴ inspiratory time,²⁷⁴ volume-limited versus pressure-limited ventilation,²⁷⁶ humidity, and size of the endotracheal tube. Placement of a jet nebulizer in the inspiratory limb of a humidified pediatric ventilator circuit 15 cm from the Y-piece had a lower efficiency of drug delivery compared to placement between the humidifier and ventilator.¹⁶⁷ A vibrating mesh nebulizer achieved twofold to fourfold higher drug delivery than the jet nebulizer.¹⁶⁷ The highest efficiency of drug delivery occurred with the vibrating mesh nebulizer placed at the inlet of the humidifier and operated continuously with a bias flow of 2 L/min in the ventilator circuit. Higher bias flows (5 L/min) reduced the efficiency of drug delivery.¹⁶⁷ In contrast, Turpenin et al found that drug delivery was improved when the nebulizer was placed at the endotracheal tube compared to placement in the inspiratory limb.²⁸¹ Moreover, as expected, breath-synchronized nebulization resulted in higher drug delivery than continuous nebulization.²⁸² The very low efficiency of nebulizers during infant mechanical ventilation, however, means that pulmonary drug deposition would be marginally influenced by altering the ventilator conditions. In fact, underdosing is a concern when a jet nebulizer is employed with settings that have very poor efficiency (as low as approximately 0.1%) for drug delivery.

Metered-Dose Inhalers versus Nebulizers

Few investigators have compared the pMDI to nebulizer efficiency during infant ventilation.^{273–276} With optimal technique, the efficiency of pMDIs (approximately 14%) is considerably higher than that of nebulizers (approximately 2%).

In Vitro Studies of Aerosol Drug Delivery in Older Children

In older children, the efficiency of pMDIs and nebulizers improve with increasing age,^{270,271} and the factors influencing drug delivery are similar to those in adults.¹⁷⁷

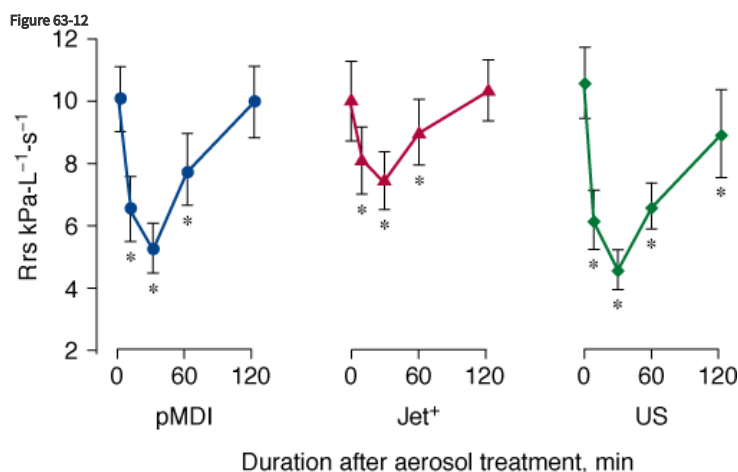
In Vivo Studies of Drug Delivery in Infants and Neonates

The lung deposition of radiolabeled aerosols in ventilated infants with bronchopulmonary dysplasia was as low as $0.98 \pm 0.2\%$ and $0.22 \pm 0.1\%$ with a pMDI and spacer or a jet nebulizer, respectively.²⁸³ Minocchieri et al employed a premature nose–throat model to demonstrate that lung deposition was reduced from $61.8 \pm 5\%$ to $26.0 \pm 1.5\%$ and $9.0 \pm 0.8\%$ of the nominal dose at continuous flows of 1, 5, and 10 L/min, respectively.²⁸⁴ Moreover, placement of the nebulizer closer to the endotracheal tube did not improve the low efficiency of aerosol delivery.²⁸⁵ In macaque monkeys, pulmonary deposition of ^{99m}Tc diethylenetriamine pentaacetate with a vibrating mesh nebulizer (Aeroneb Pro prototype) was more than 20-fold higher than that with a jet nebulizer (MistyNeb, Airlife Inc., Mountain View, CA).²⁸⁶

Clinical Studies

The low efficiency of drug delivery is offset by the smaller size of the lung in infants and children. Compared to a standard 1.3 mcg of albuterol/kg body weight delivered by each actuation from a Ventolin-HFA pMDI in a 70-kg, spontaneously breathing, adult male, the range for spontaneously breathing infants and small children varies from 0.8 to 2.6 mcg of albuterol delivery/kg body weight/actuation with a HFA-pMDI and AeroChamber Plus valved holding chamber.²⁸⁷ Similar considerations apply for infants and children receiving mechanical ventilation.

Inhaled bronchodilators (β -adrenergic and anticholinergic drugs)^{6–10,288–292} are effective in ventilated neonates and infants with acute, subacute, and chronic lung disease. Response to inhaled albuterol was similar to that seen after intravenous administration.²⁸⁹ Albuterol administered with a pMDI and AeroChamber was found to be more effective than administration with a nebulizer (Fig. 63-12).^{10,291} Ultrasonic nebulizers may be as (or more) effective than pMDIs for bronchodilator administration in infants (Fig. 63-12). In addition to monitoring airway resistance and pulmonary compliance, changes in heart rate should be assessed to prevent overdosing. ICSs have been advocated in infants for prevention^{293,294} and treatment of bronchopulmonary dysplasia,^{295–297} but their efficacy has not been established. In summary, during infant mechanical ventilation, pMDIs appear to be more efficient than jet nebulizers; with a careful technique of administration, one or two doses from an albuterol pMDI and spacer chamber should suffice for routine therapy. The aerosol delivery efficiency of vibrating mesh nebulizers is comparable to pMDIs, but the clinical effectiveness of these devices in infants and children needs further study.



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Total resistance of the respiratory system (R_{rs}) before and at 15, 30, 60, and 120 minutes after albuterol via pressurized metered-dose inhaler (pMDI), jet nebulizer (Jet), and ultrasonic nebulizer (US) during infant ventilation. *, posttreatment R_{rs} was significantly lower than pretreatment R_{rs} ($p < 0.0001$). The response to albuterol after administration with a pMDI was greater than that obtained with a jet nebulizer but was comparable to that obtained with an ultrasonic nebulizer. (Used, with permission, from Fok et al.²⁹¹)

Techniques of Aerosol Administration

The optimal techniques for administration of inhaled drugs to ventilated patients are based on the various factors discussed above. The technique employed may have to compromise between the optimum operating characteristics of the device and the patient's condition. For example, although a higher duty cycle increases aerosol delivery,^{195,199} it may worsen dynamic hyperinflation in patients with airflow limitation.²²⁵ With this caveat in

mind, the technique of administration for pMDIs in ventilated patients is shown in Table 63-4 and that for nebulizers in Table 63-5. The key aspects of the technique with pMDIs are to place an appropriate adapter at a short distance (approximately 15 cm) from the endotracheal tube and to synchronize actuation with inspiratory flow. For nebulizers, it is critical to use the device as recommended by the manufacturer and to place it at a distance from the patient. When these techniques are employed, adequate drug deposition is achieved in the lung^{175,195,200} and a significant bronchodilator response is observed.^{12,21,213,215,224,248,250–255} Most ventilated patients, however, do not show incremental effects with higher drug doses.²¹⁵

Table 63-4: Optimal Technique for Drug Delivery by pMDI in Ventilated Patients

1. Review order, identify patient, and assess need for bronchodilator.
 2. Suction endotracheal tube and airway secretions.
 3. Shake pMDI and warm to hand temperature.
 4. Place pMDI in spacer chamber adapter in ventilator circuit.
 5. Remove HME. Do not disconnect humidifier.
 6. Coordinate pMDI actuation with beginning of inspiration.
 7. Wait at least 15 seconds between actuations; administer total dose.
 8. Monitor for adverse response.
 9. Reconnect HME.
 10. Document clinical outcome.

Abbreviations: HME, heat and moisture exchanger; pMDI, pressurized metered-dose inhaler.

Table 63-5: Optimal Technique for Drug Delivery by Jet Nebulizer in Ventilated Patients

1. Review order, identify patient, and assess need for bronchodilator.
 2. Suction endotracheal and airway secretions.
 3. Place drug in nebulizer to fill volume of 4 to 6 mL.
 4. Place nebulizer in the inspiratory line 18 in (46 cm) from the patient Y connector.
 5. Turn off flow-by or continuous flow during nebulizer operation.
 6. Remove HME from circuit (do not disconnect humidifier).
 7. Set gas flow to nebulizer at 6 to 8 L/min.

a. Use a ventilator if it meets the nebulizer flow requirements and cycles on inspiration, *or*

b. Use a continuous flow from external source.
 8. Adjust ventilator volume or pressure limit to compensate for added flow.
 9. Tap nebulizer periodically until nebulizer begins to sputter.
 10. Remove nebulizer from circuit, rinse with sterile water and run dry, store in safe place.
 11. Reconnect humidifier or HME, return ventilator settings and alarms to previous values.
 12. Monitor patient for adverse response.
 13. Assess outcome and document findings.

Abbreviation: HME, heat and moisture exchanger.

Use of Metered-Dose Inhalers or Nebulizers

Nebulizers and pMDIs delivered an equivalent mass of aerosol beyond the endotracheal tube in a ventilator model;²⁰² in ventilated patients, both devices produced similar therapeutic effects.^{246,252,254} When used optimally, pMDIs and nebulizers are equally effective in the treatment of patients with obstructive lung disease.⁶²

pMDIs are preferred in ventilated patients because of several problems associated with the use of nebulizers.^{155–157,226,232,233} Before using a nebulizer in a ventilated patient, it is imperative to characterize its efficiency in a ventilator circuit, under the typical clinical conditions in which it will be employed.

Unless scrupulously cleaned and disinfected, nebulizers can be a source for aerosolization of bacteria,²⁹⁸ and thus predispose patients to nosocomial pneumonia.²⁹⁹ The gas flow driving the nebulizer produces additional airflow in the ventilator circuit, but this is compensated for in most modern ventilators. In contrast, pMDIs are easy to administer, require less personnel time, provide a reliable dose, and do not pose a risk of bacterial contamination. Use of pMDIs rather than nebulizers saves cost and time.^{300–302} Costs of bronchodilators have a significant impact on overall costs of care.³⁰²

In summary, pMDIs offer several advantages over nebulizers for routine bronchodilator therapy. One survey found that most reporting centers (57%) were using pMDIs in neonates and their use had steadily increased since 1988;³⁰³ similar trends are observed in adults.

Drug Toxicity

Higher doses of β -agonists delivered by a pMDI can cause atrial and ventricular arrhythmias.^{304–306} Manthous et al²¹³ observed sinus tachycardia or supraventricular ectopy after a cumulative dose of 7.5 mg of albuterol administered by a nebulizer in four of ten patients, and most of the remaining patients developed premature atrial and ventricular contractions with a cumulative dose of 15 mg.²¹³ Although most investigators have reported no adverse effects following administration of albuterol with a pMDI,^{12,129,194,213,214} a dose-dependent increase in heart rate occurs with higher doses.²¹⁵ No significant arrhythmias or other serious cardiovascular side effects were observed in ambulatory patients in an emergency department who were treated for acute asthma with up to 16 puffs each of albuterol or fenoterol administered with a pMDI attached to a holding chamber and face mask.⁶⁵ Continuous nebulization of β -agonists is effective and safe in nonintubated children³⁰⁷ and adults⁷⁴ with acute severe asthma, but the efficacy and safety of continuous nebulization has not been established in ventilated patients. A few anecdotal reports have described cardiotoxicity secondary to CFCs.³⁰⁸ Adverse cardiac effects are unlikely to occur with the doses recommended in clinical practice, particularly if there is a short interval between successive doses, because CFCs have a short half-life (<40 seconds) in blood after administration via pMDI in healthy volunteers.³⁰⁹ With a catheter connected to a pMDI nozzle, however, a substantial portion of the total mass output of the pMDI is delivered directly onto the tracheobronchial mucosa and there is potential for local and systemic toxicity.²³⁸

Arterial oxygen tension (PaO_2) may decrease transiently after β -agonist inhalation, probably secondary to pulmonary vasodilation mediated by β_2 -receptors.^{310,311} In contrast, reduction in PaO_2 was either not seen after inhalation of antimuscarinic agents³¹² or was less than the decrement produced by albuterol and salmeterol.⁴³ The decrease in PaO_2 with albuterol inhalation is transient and small; hence, its clinical significance is doubtful.⁴³ Significant drops in PaO_2 , however, can occur occasionally,^{311,313} and close monitoring may be required in patients with marginal oxygenation.

Blood glucose and insulin levels increase after β -agonist administration, whereas serum potassium levels decrease.^{310,314} The clinical importance of these effects is inconclusive; however, in patients with severe airway obstruction fall in serum potassium in the presence of hypoxia has the potential to cause cardiac arrhythmias. A variety of central nervous system effects including anxiety, nervousness, tremor, irritability, insomnia and headaches are associated with use of β -adrenergic agents.

Guidelines for Inhaled Therapy

Subtle differences in the method of administration can markedly decrease aerosol deposition in the lower respiratory tract.^{175,195,202,208,213,226} Routine bronchodilator therapy can be given successfully with assisted modes if aerosol delivery is synchronized with inspiratory flow from the ventilator. Based on the recommendation for use of pMDIs in ambulatory patients, some investigators use a postinspiratory breathhold after aerosol administration.^{129,246} With an optimal technique, this maneuver does not influence bronchodilator response in ventilated patients.²⁵⁰ Mouloudi et al^{251,253} have also shown that increasing the tidal volume or using pressure-control (decelerating flow pattern) versus volume-control (constant flow pattern) ventilation did not affect the response to albuterol inhalation with a pMDI. These findings are consistent with other studies showing shallow dose-response curves in ventilated patients.^{9,129,215,245,254} Thus, there may be little justification for manipulating ventilator settings to enhance drug deposition in the lung provided an optimal pMDI technique is employed.

Although humidification of the circuit reduces aerosol deposition by approximately 40%,^{195,197} bypassing the humidifier is not routinely recommended because it would require disconnection of the ventilator circuit and several minutes would be added to each bronchodilator treatment while waiting for the circuit to become dry. Moreover, when heated wire circuits are employed, there was no reduction in delivery efficiency of albuterol with a pMDI and spacer for up to 60 minutes after turning on the heated humidifier,³¹⁵ and after 3 hours of humidifier operation, it took 10 minutes after shutting off the humidifier for aerosol delivery to improve. Even with a humidified circuit, a significant effect is observed with as few as 4 puffs.²¹⁵ Bypassing the

humidifier may be one method to improve pulmonary drug delivery with more expensive, nonbronchodilator agents. Before such a practice can be routinely recommended, however, studies are needed to document the safety of repeatedly administering nonhumidified gases, albeit for limited periods, to ventilated patients.

Bronchodilator Therapy during Noninvasive Ventilation

Noninvasive positive-pressure ventilation (NIPPV) is being increasingly employed for treatment of patients with acute and chronic respiratory failure.³¹⁶ Successful application of NIPPV with a nasal or face mask in patients with acute exacerbations of COPD can often obviate the need for endotracheal intubation and improve mortality.^{317–320} Patients with acute or acute-on-chronic respiratory failure who are receiving NIPPV often require inhaled bronchodilators for relief of airway obstruction. One option is to remove the patient from NIPPV and administer bronchodilators by pMDI or nebulizer with a face mask as patients can tolerate brief periods of discontinuation needed for bronchodilator therapy.³²¹ It is preferable, however, to continue NIPPV without interruption, especially in acutely hypoxic or dyspneic patients.

The delivery of aerosols during NIPPV is complicated by high inspiratory flows that increase turbulence and the associated high inertial forces cause greater particle impaction in central airways.^{223,322} On the other hand, application of positive pressure increases tidal volume and reduces respiratory rate, both of which tend to enhance aerosol delivery.³²³ Moreover, increase in expiratory time because of a slower respiratory rate could enhance particle sedimentation and alter the pattern of drug deposition during exhalation.³²³ Initial enthusiasm for administration of bronchodilators with intermittent positive-pressure breathing³²⁴ was generated when it was shown that higher plasma levels of albuterol were achieved by this technique compared to administration by pMDI alone. Later studies, however, found that drug delivery to patients with severe COPD and relatively fixed airway obstruction was reduced by administration with intermittent positive-pressure breathing.³²⁵

The use of continuous positive airway pressure (CPAP) may be more helpful than intermittent positive-pressure breathing in reducing work of breathing in patients with acute bronchoconstriction.^{326,327} However, in a bench model, 10-cm H₂O CPAP reduced drug delivery from a jet nebulizer.³²⁸ Mask CPAP is optimally employed with a continuous fresh gas flow that approaches or exceeds the patient's inspiratory flow rate so as to minimize the additional work imposed by a demand valve system.³²⁹ Entrainment of aerosols into the high rates of gas flow used during CPAP dilutes the drug and favors evaporation leading to reduced droplet size, cooling, and hypertonicity,¹⁵² with the potential to produce paradoxical bronchoconstriction. Likewise Chatmongkolchart et al found that application of CPAP alone, especially higher levels of CPAP, reduced aerosol delivery.³³⁰

Noninvasive pressure support provides a more sustained boost to inspiratory effort than intermittent positive-pressure breathing by maintaining a constant preset airway pressure.³³¹ Application of pressure support reduces inspiratory muscle effort and respiratory rate, increases tidal volume and improves arterial blood-gas values.^{332,333} Pressure support could also produce more uniform aerosol deposition in the lungs by reversing small airway closure and microatelectasis.³³⁴

During bilevel ventilation, Faroux et al³³⁵ found that in children with cystic fibrosis, pulmonary deposition of radiolabeled aerosol was increased by the application of 10 cm H₂O pressure support. This study, however, employed a homecare ventilator that does not incorporate a leak port in the circuit and the nebulizer was active only during inspiration. Chatmongkolchart et al³³⁰ found a fivefold variation (between 5% and 25% of the nominal dose) in the amount of albuterol delivered by a jet nebulizer in vitro; delivery was highest (25%) when the nebulizer was close to the patient (between the leak port and patient connection), inspiratory pressure was high (20 cm H₂O), and expiratory pressure was low (5 cm H₂O).³³⁰ Although nebulizer operation during bilevel positive-pressure ventilation does not influence ventilator function,³³⁰ nebulizer operation with bilevel positive-pressure ventilation enhances drug delivery and produces a finer aerosol with a higher proportion of respirable particles compared to nebulizer operation alone.³³⁶ NIPPV, however, did not improve radioaerosol deposition compared to spontaneous breathing in healthy volunteers.³³⁷

Intrapulmonary percussive ventilation was found to produce a finer aerosol than standard jet nebulization, but pulmonary deposition of radiolabel was much more variable and unpredictable with this technique in healthy volunteers compared to standard nebulizer therapy, and it was not recommended as a mode of aerosol delivery.³³⁸ The use of high-frequency oscillation for radioaerosol delivery was found to reduce total pulmonary drug deposition in normal subjects, and in patients with chronic airflow obstruction, despite a reduction in the size of the aerosol.³³⁹ In normal subjects, the regional distribution of aerosol deposition was unchanged, but in the patients with airflow obstruction a larger proportion of total aerosol deposition occurred in the peripheral lung during high-flow oscillation.³³⁹

The optimum settings required for maximum drug delivery with a pMDI during NIPPV have not been established. Significant bronchodilator responses occur after albuterol administration with a jet nebulizer or a pMDI in stable patients receiving NIPPV with face mask.^{340,341} Nava et al used a homecare ventilator and volume-assured pressure support with dry, unwarmed gas to obtain a guaranteed tidal volume of 10 mL/kg in clinically stable patients

with severe COPD.³⁴⁰ With the pMDI and spacer placed in the inspiratory limb of the ventilator circuit immediately after the patient Y, they found a higher change in FEV₁ after albuterol administration compared to placebo but the effect was comparable to that obtained with a pMDI and spacer during spontaneous breathing.³⁴⁰ Similarly, Parkes and Bersten employed CPAP at 10 cm H₂O and used nonhumidified compressed gas at a flow rate of 50 L/min with a tight face mask fit in nine stable patients with asthma.³²⁸ Despite reduction in total aerosol delivery, application of CPAP did not affect the bronchodilator response to nebulized albuterol.³²⁸ Patients with acute asthma exacerbations who received albuterol administered during bilevel positive airway pressure ventilation³⁴¹ had more rapid and greater improvement in peak expiratory flow than did patients receiving a similar dose by nebulizer alone. It is unclear, however, if this was caused by relief provided by pressure support, or enhanced drug delivery or drug effect produced by NIPPV, or a combination of the two. Both pMDIs and nebulizers could be successfully employed during NIPPV by using the techniques recommended by Hess.³⁴²

The position of the leak port in the single-arm circuit used for bilevel positive-pressure ventilation influences nebulizer efficiency, but it does not influence pMDI efficiency of drug delivery.³⁴³ Nebulizer efficiency is higher with the leak port in the circuit as compared to a leak port in the face mask.³⁴³ Calvert et al reported that a nebulizer placed between the leak port and ventilator performs with a higher efficiency than placement of the nebulizer between the leak port and face mask.³³⁶ In contrast, Abdelrahim et al found a higher drug delivery with the nebulizer placed between the leak port and face mask than with the nebulizer placed between the ventilator and leak port.³⁴⁴ The difference in the results of the two studies^{336,344} may be explained by the fact that the former investigators placed the nebulizer at a farther distance from the leak port than did the latter investigators. Moreover, drug delivery with a vibrating mesh nebulizer at either position of the nebulizer was more than twice that delivered by a conventional jet nebulizer.³⁴⁴

In summary, delivery of aerosols to patients receiving NIPPV is extraordinarily complex. Several factors related to the type of ventilator employed, the type of aerosol generator and its position in the ventilator circuit, the mode of ventilation, tidal volume, inspiratory flow rate, inspiratory and expiratory pressure settings, circuit humidity, the density of the gas in the circuit, type of mask interface, mask size, and patient characteristics all influence the efficiency of aerosol delivery in this setting. Moreover, the single-circuit design, continuous air flow, and the presence of a leak port further complicate aerosol delivery.

Important Unknowns

Despite frequent use of bronchodilators in ventilated patients, their effects on clinically relevant outcomes, such as duration of mechanical ventilation, length of ICU stay, development of ventilator-associated pneumonia, and mortality have not been studied. In fact, few randomized controlled studies have been performed in this patient population.⁶² Variations in the methods of administration and assessment of response make it difficult to compare results between various studies.

Further work is also needed to determine whether newer generation of vibrating mesh nebulizers, soft mist inhalers, and dry powder inhalers can be successfully adapted for use during mechanical ventilation.

The Future

Ventilated patients present a unique opportunity to exploit the advantages of the inhaled route for drug delivery. The past two decades have seen impressive gains in knowledge and understanding of methods to deliver inhaled therapies to ventilated patients. Within the past few years, newer aerosol-generating devices designed for use during mechanical ventilation have become available. These devices are far more efficient than previous jet nebulizers. In the future, further improvements in device design will result in higher efficiency devices that also ensure precision, reliability, and consistency of dosing.

Some modern ventilators have integrated aerosol-generating devices, and this integration should further improve the characterization of aerosols and consistency of drug dosing.

NIPPV is being increasingly employed, especially in patients with acute exacerbations of COPD. Further work is needed to improve our understanding of the factors influencing inhaled drug delivery in this setting. Delivery devices that are designed specifically for mask ventilation would be helpful. Moreover, recent gains in improving aerosol delivery to infants and children will lead to greater clinical application of inhaled therapies in this population.

Summary and Conclusion

Bronchodilator therapy is commonly employed in ventilated patients. Inhaled β -agonist and anticholinergic bronchodilators are widely used, whereas use of theophylline in the ICU has declined significantly. Although ICSs are being employed in ventilated patients, especially in combination with long-

acting β -agonists, further investigations are needed to determine the optimal methods of administration, drug dosage, and efficacy; and to assess their risk-to-benefit ratio in this setting. Optimal techniques for employing pMDIs and nebulizers in ventilator circuits have been developed as a result of better understanding of the factors influencing aerosol delivery to the lower respiratory tract of ventilated patients. Important variables that influence aerosol delivery include the type of nebulizer used and its position in the circuit, actuation of a pMDI into an inline chamber spacer, timing of actuation, ventilator mode, tidal volume, circuit humidification, and duty cycle. With proper technique, drug deposition in the lower respiratory tract of ventilator-dependent patients is comparable to that in ambulatory patients. A somewhat higher dose than that used in ambulatory patients is recommended in ventilated patients, mostly to compensate for the effects of humidity in the ventilator circuit. Typically, dose–response curves to bronchodilators in ventilator-supported patients are shallow and the duration of the drug effect is variable. Although both nebulizers and pMDIs are employed for administering bronchodilators in mechanically ventilated patients, pMDIs offer several advantages over nebulizers for routine bronchodilator therapy.

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